



MONASH University

FIT5152 User interface design and usability - S2 2025
Submission 3 - High-Fidelity Prototype and Report

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Part 1: Figma Link

<https://www.figma.com/design/1safprRUidKkz0urUckjSf/Donner---A-Donation-App--Community-?node-id=982-2757&t=CBiXvMM38L0LdmG1-1>

Part 2: Written Report

Introduction

This report presents the high-fidelity prototype developed for Toy Trove, a mobile application designed to address key user needs identified through comprehensive research and user-centered design processes. Building upon the personas and user stories, this prototype implements selected user requirements through an interactive Figma design. The prototype demonstrates the practical application of established usability principles, design guidelines, and accessibility standards to create an intuitive and inclusive user experience. This document provides a detailed examination of the prototype's screens, explaining the implementation of design theory and evaluating usability through heuristic analysis, showcasing how user-centred design principles have been translated into a functional, accessible interface.

User Requirements

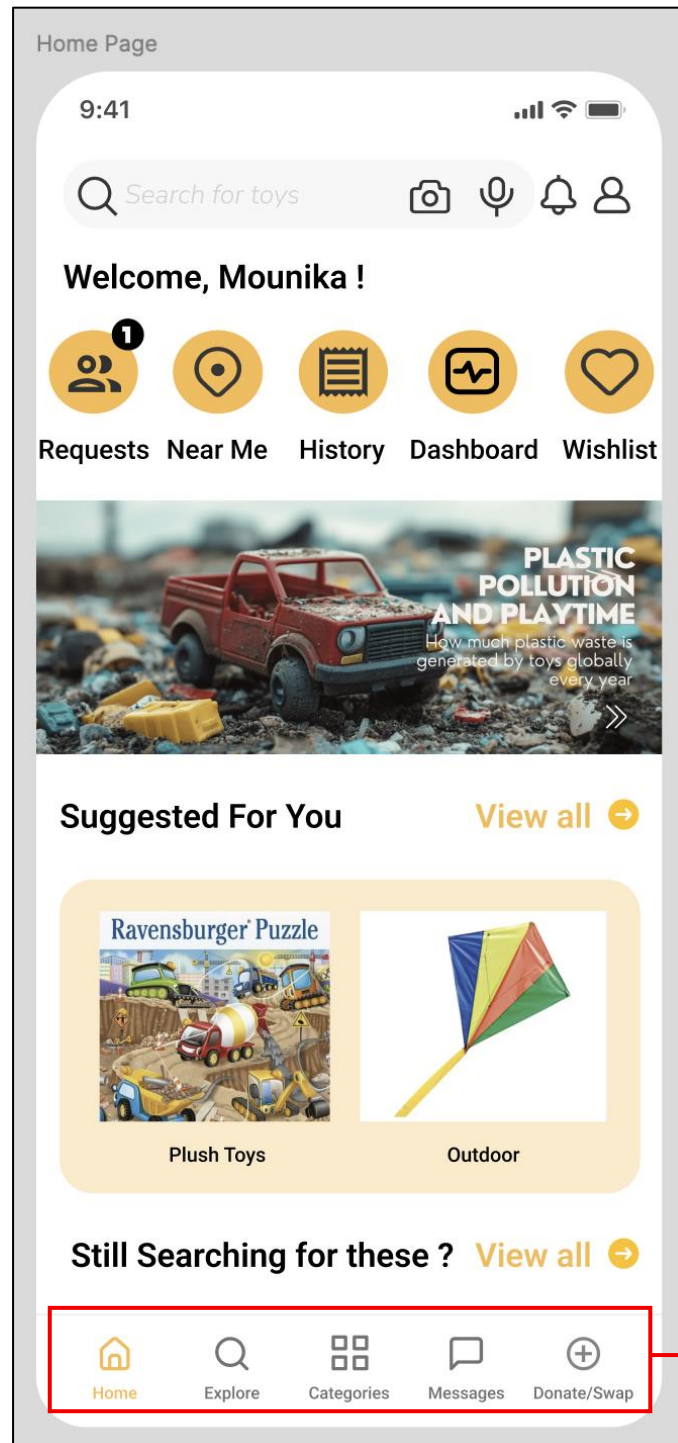
1. Users should be able to access key actions (such as finding, donating, and swapping items) through clearly labelled navigation options.
2. Users should be able to access and manage a personal dashboard that tracks all toys they have shared, swapped, or received.
3. Users should be able to communicate with others through secure, in-app messaging that doesn't expose personal contact details
4. Users should be able to browse and filter available items using detailed condition tags selected from a standardized checklist
5. Users should be able to view verified cleanliness and safety information for every toy before requesting or swapping.
6. Users should be able to contribute toys in bulk with clear details about quantity, condition, and age suitability.

Prototype Screens – Toy Trove

Screen 1 – Homepage

User Requirement:

Users should be able to access key actions (such as finding, donating, and swapping items) through clearly labelled navigation options.



Navigation Type:
Bottom Tab Bar

Figure 1:Homepage(Before scrolling)

Design Guideline:

Navigation Type: Bottom Tab Bar

This home page implements a **persistent bottom tab bar navigation**, which is a mobile design best practice for primary navigation. The tab bar contains 5 options (Home, Explore, Categories, Messages, Donate/Swap), positioned at the bottom for thumb-reachable access on larger mobile screens. This navigation type remains globally available and persistent across screens, providing users with a sense of comfort and allowing them to quickly reset their navigation path or change their goal. The implementation follows the guideline of using 3-5 most important options for primary tasks, ensuring the navigation feels natural and reduces cognitive load by keeping essential actions constantly visible and accessible.

Screen 2 – Inbox

User Requirement:

Users should be able to communicate with others through secure, in-app messaging that doesn't expose personal contact details.

Design Principle:

Common Region (Gestalt Principle):

This inbox screen effectively applies the **Common Region gestalt principle** from visual design theory. Each conversation is contained within a subtle white background card boundary that separates it from other messages. According to gestalt psychology (Tidwell et al., 2020), objects located within the same boundary are perceived as a singular group, working in conjunction with proximity. Each message card groups multiple elements (profile picture, sender name, message preview, timestamp, unread badge) into a unified, identifiable object that users can immediately recognize as a complete conversation item. This boundary-based grouping allows users to quickly scan their inbox and perceive each conversation as a distinct, tappable unit rather than a cluttered collection of disconnected text elements.

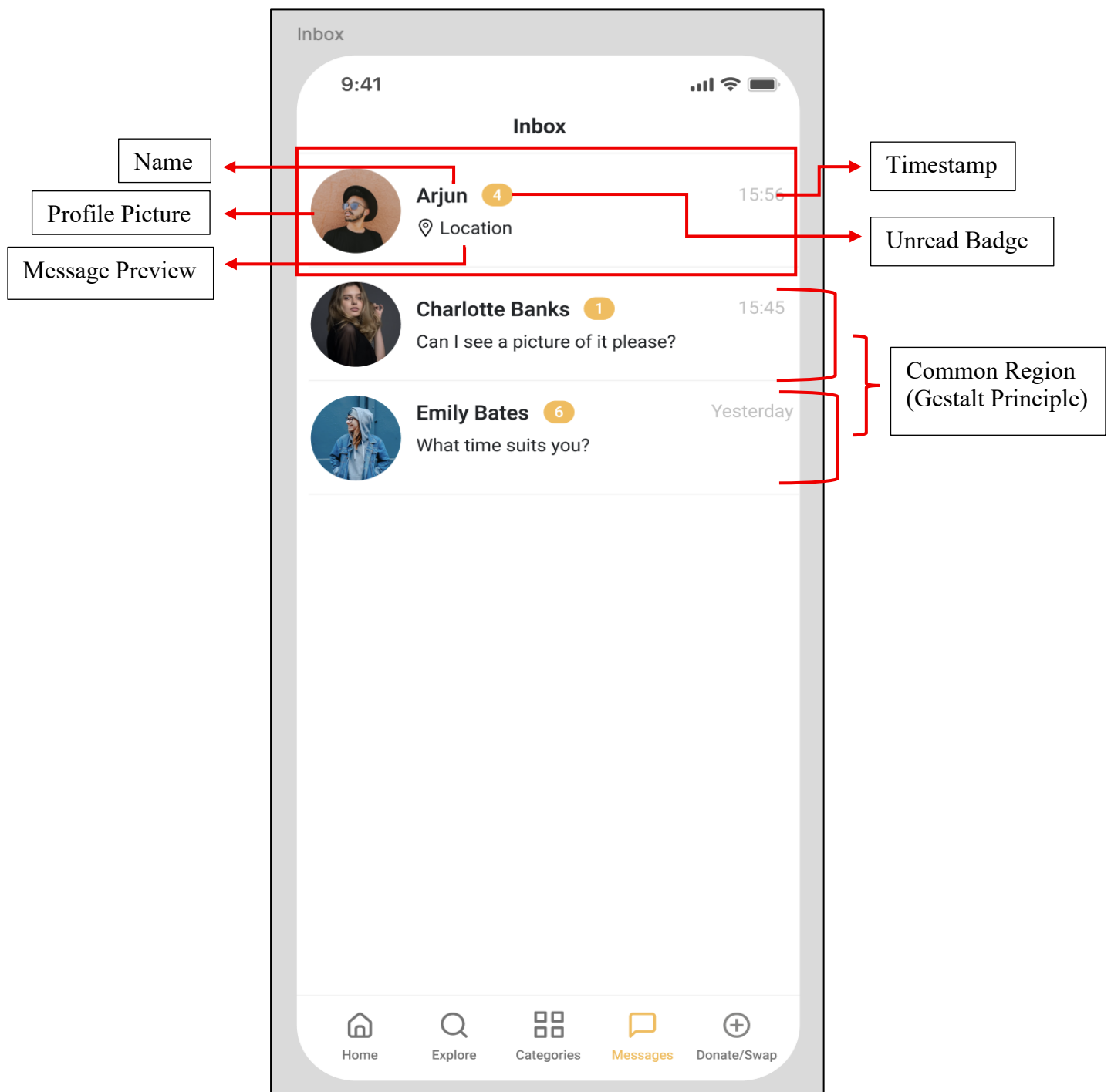


Figure 2: Inbox

Screen 3 – Chat

User Requirement:

Users should be able to communicate with others through secure, in-app messaging that doesn't expose personal contact details.

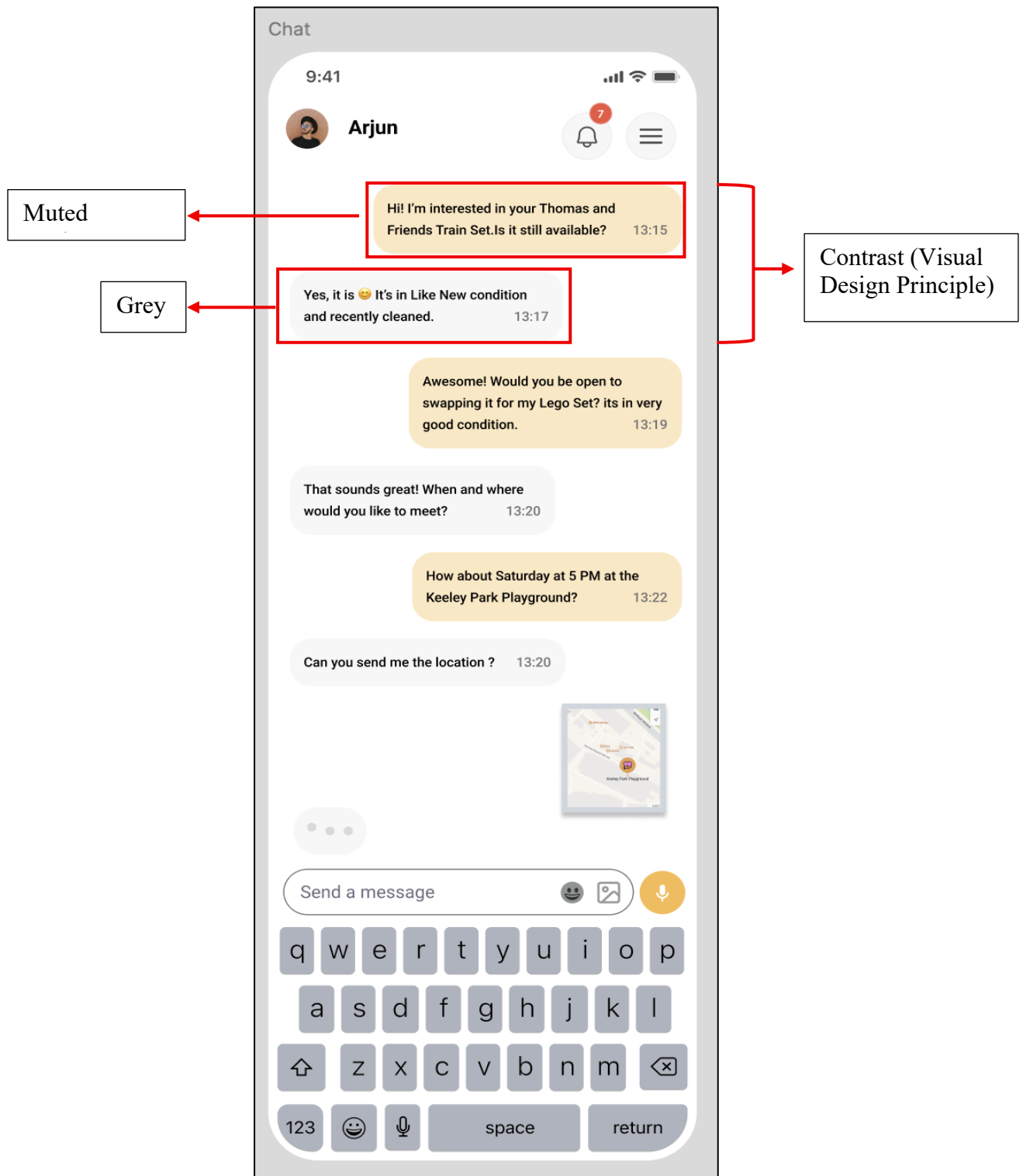


Figure 3: Chat

Design Principle: Contrast (Visual Design Principle):

This chat screen effectively applies the **Contrast** principle from visual design theory (Tidwell et al., 2020) to distinguish between conversation participants. Contrast refers to the visual difference between design elements, particularly the colour of foreground elements. The interface uses contrasting message bubble colours – muted apricot for sender messages (right-aligned) and grey for receiver messages (left-aligned) - creating immediate visual distinction. This colour contrast, combined with opposing spatial alignment, allows users to instantly identify message ownership without needing to read names. According to visual

design principles (UX Daily, 2025), effective use of contrast guides user attention and creates clear visual hierarchies, enabling users to scan conversations efficiently and understand the dialogue flow at a glance.

Screen 4 – Browse Toys

User Requirements:

- 1.Users should be able to browse and filter available items using detailed condition tags selected from a standardized checklist.
- 2.Users should be able to view verified cleanliness and safety information for every toy before borrowing or swapping.

Design Principle:

Jakob's Law (Best Practices for Interface Design):

This browse screen effectively applies **Jakob's Law** (Nielsen, 2019), which states that users prefer interfaces to work the same way as other sites they already know. The design follows the "global design template" prevalent in 2025 (Checker, 2025), employing familiar e-commerce conventions: card-based product listings resembling Amazon, Myntra, or Flipkart (Indian e-commerce apps), filter and sort buttons in expected positions, colour-coded condition tags matching marketplace patterns, and grid/list view toggles common across platforms. While critics argue such convergence creates "boring sameness" (Checker, 2025), Jakob's Law justifies this approach- data-driven decisions prove rounded buttons and familiar hierarchies reduce cognitive load (Yablonski, 2025) and improve conversion rates. This strategic use of recognizable patterns enables immediate user comprehension without instruction, demonstrating how design systems prioritize usability over novelty, ensuring accessibility across diverse user groups.

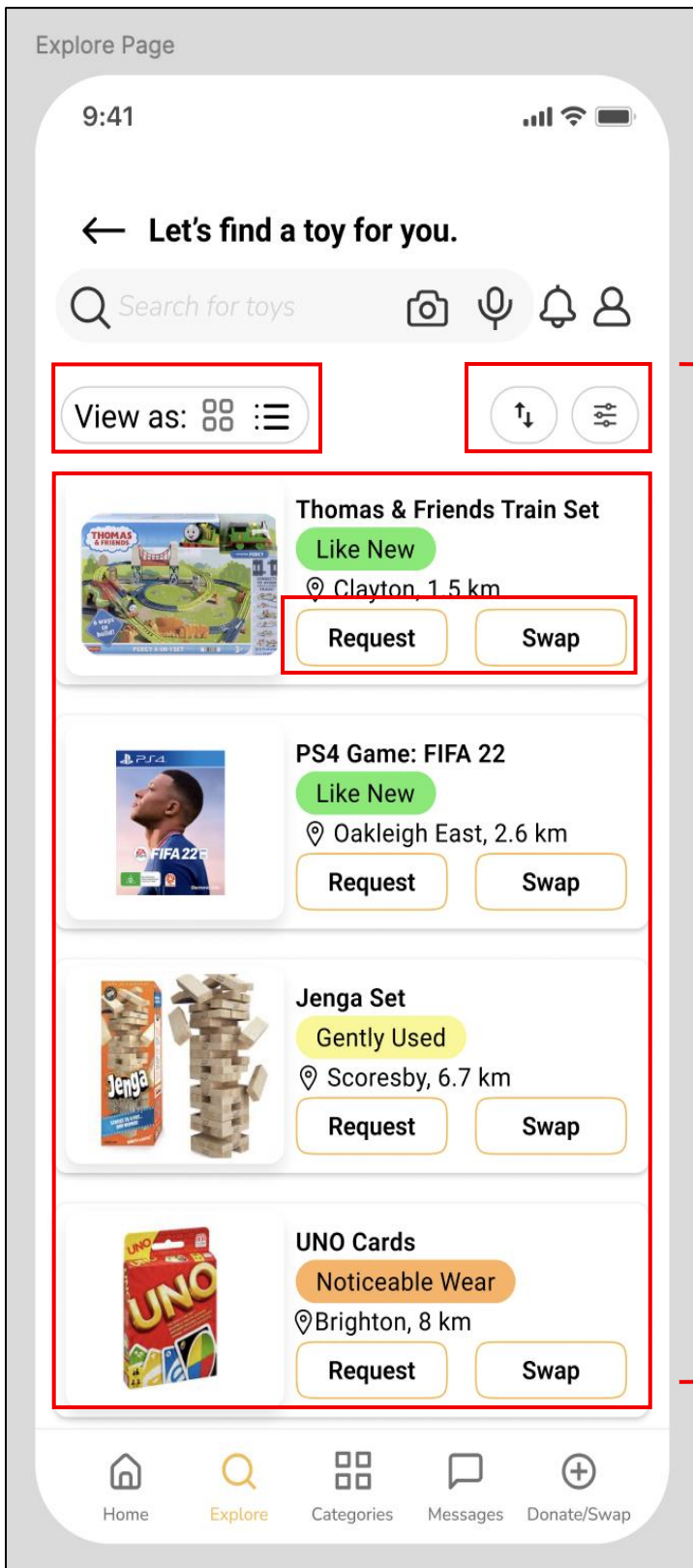


Figure 4(a): Explore toys page

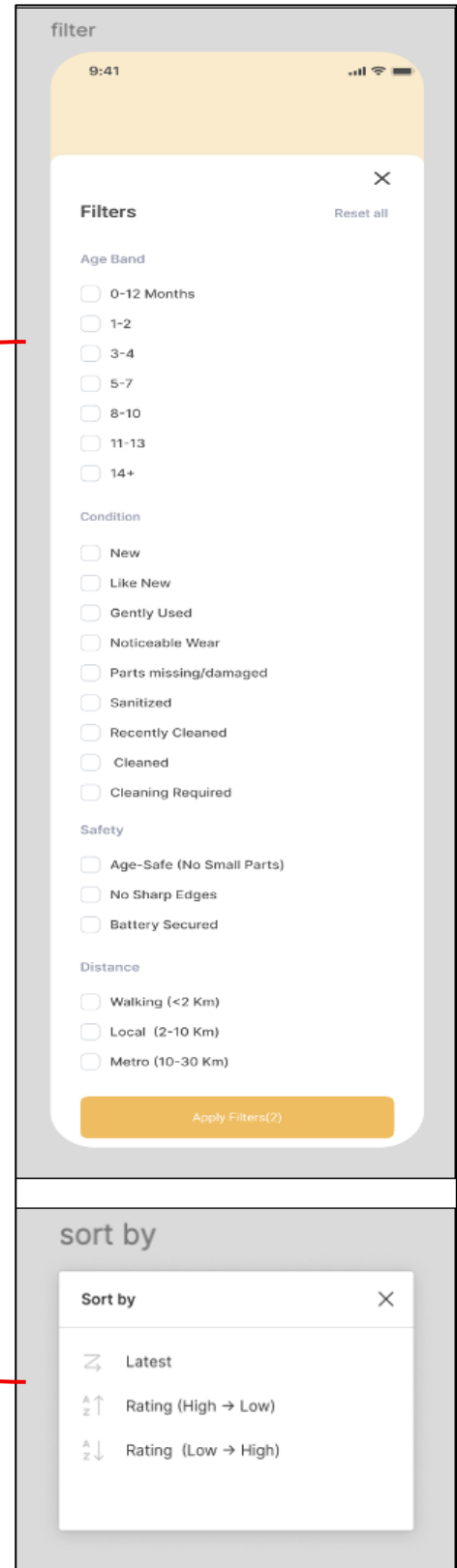
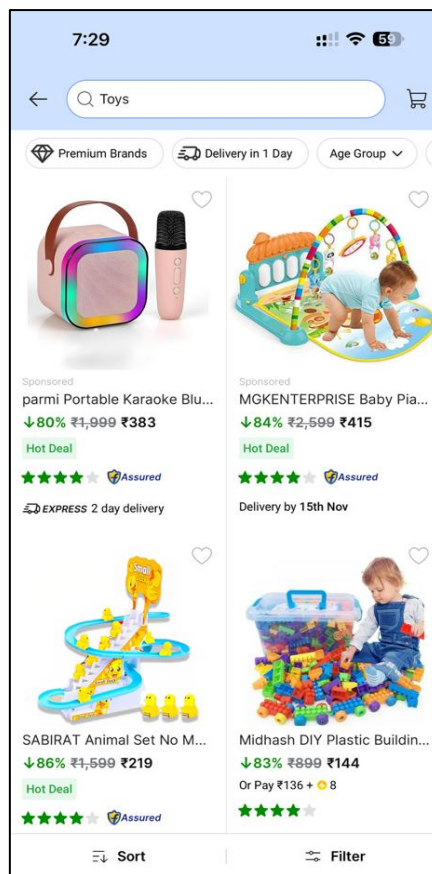
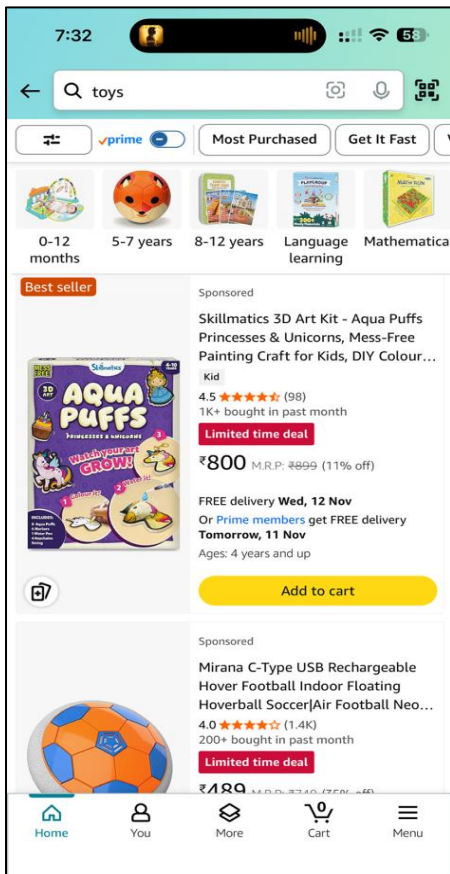


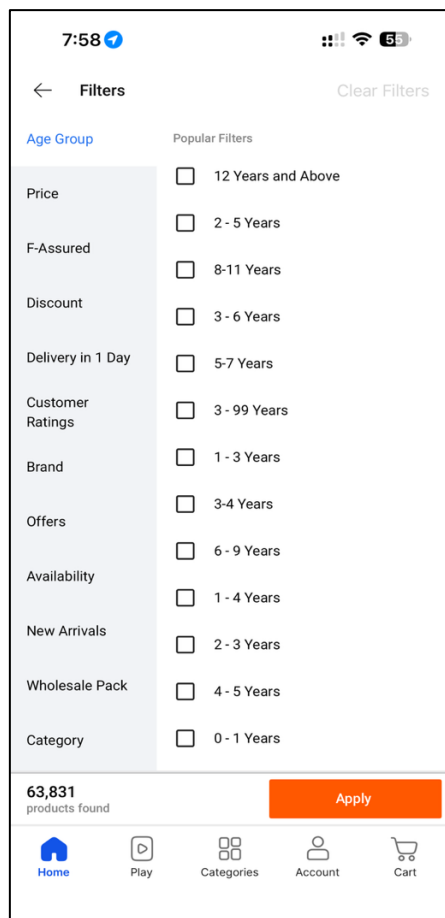
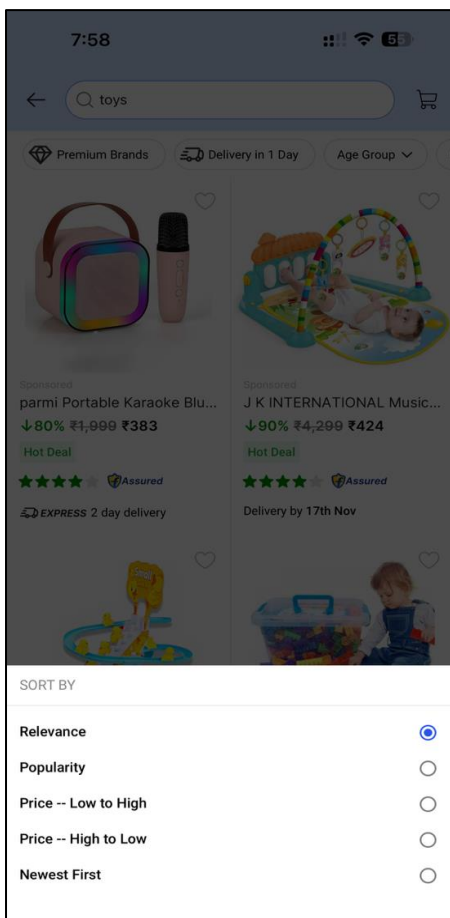
Figure 4(b): Filter and Sort by options

Jakob's Law



E-commerce platforms share remarkably similar visual designs, creating a standardized marketplace aesthetic.

Jakob's Law examples



Filter and sort options occupy standardized positions with uniform layouts throughout these platforms.

Figure 5: E-commerce apps (top left: Amazon. others: Flipkart)

Screen 5 – Product Description

User Requirement:

Users should be able to view verified cleanliness and safety information for every toy before borrowing or swapping.

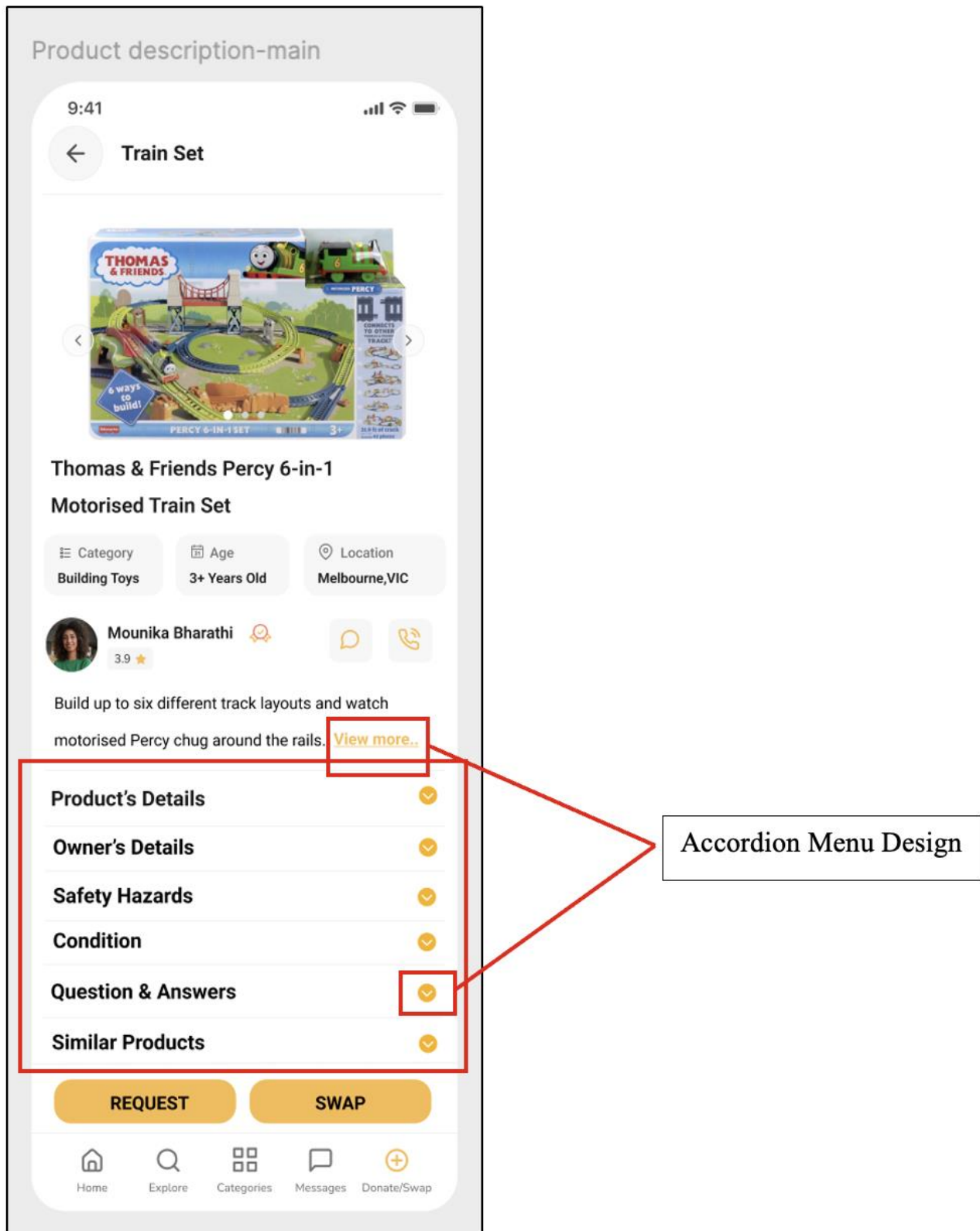


Figure 7(a): Product description

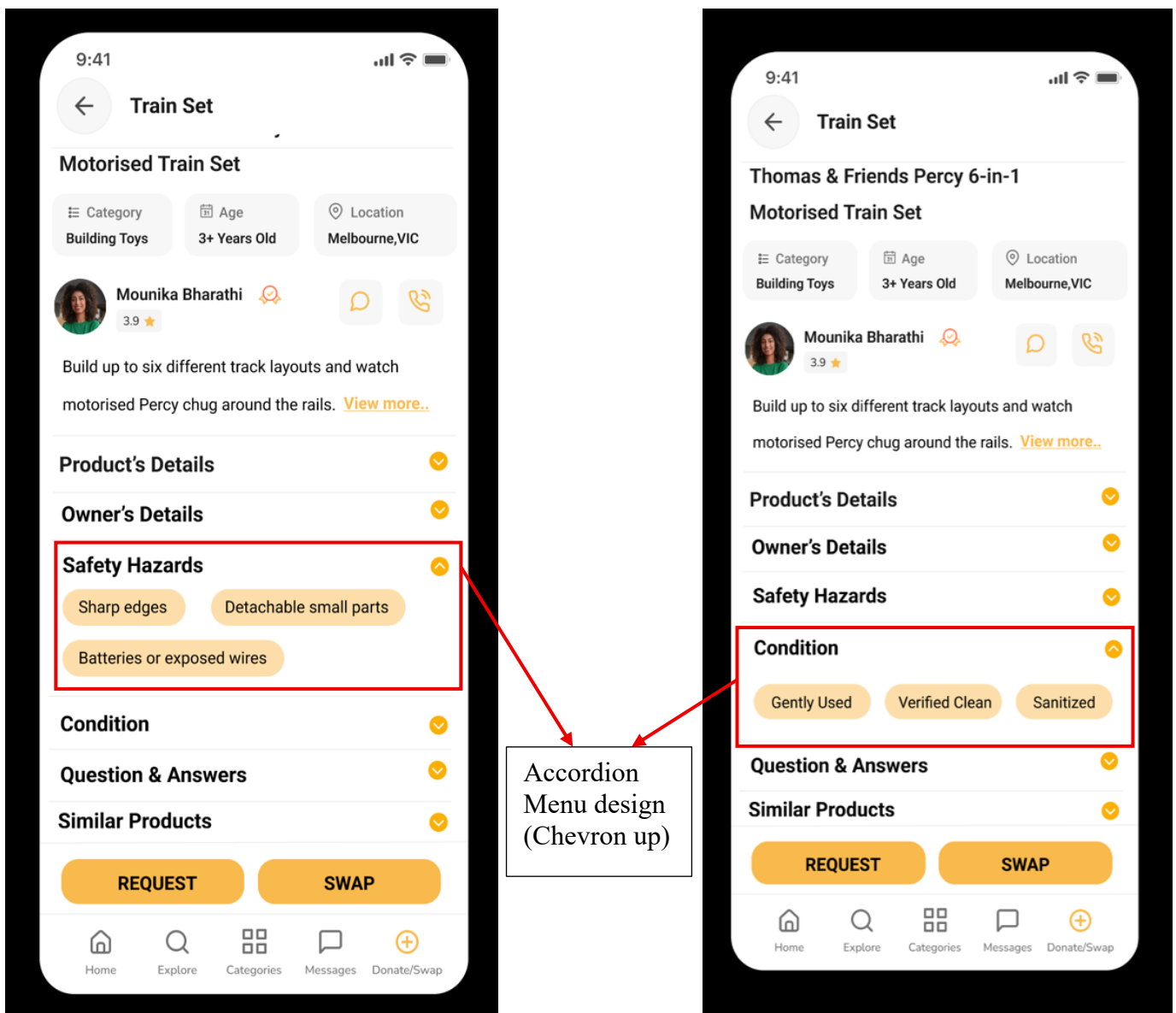


Figure 7(b): Product Description (Dropdown)

Design Principle:

Accordion Menu Design (Navigation & Menu Guidelines):

This product description screen implements **accordion menu design** from navigation and menu design guidelines (Tidwell et al., 2020). Accordions allow list menus to display content in an intuitive way where headers can be clicked to reveal information. Each section (Product's Details, Owner's Details, Safety Hazards, Condition, Question & Answers, Similar Products) uses clear chevron icon signifiers providing both affordance and natural mapping - down chevrons indicate collapsed content, up chevrons show expanded sections. This design supports gradual disclosure of complex product information, preventing cognitive overload by showing only essential headers initially. Users can selectively expand relevant sections, making the interface scannable while maintaining access to comprehensive toy verification details including cleanliness and safety information critical for informed borrowing decisions.

Screen 6 – Dashboard

User Requirement:

Users should be able to access and manage a personal dashboard that tracks all toys they have shared, swapped, or received.

Design Guideline:

Menu Organization and Sequencing (Navigation and Menu Design):

This profile overlay effectively implements **menu organization and sequencing** principles for efficient browsing. According to navigation design guidelines, organizational groupings must have understandable relationships. The screen uses ambiguous organization based on subjective rules, grouping menu items by meaningful thematic associations: personal settings (Edit profile, Notifications, Language), app configuration (Security, Theme), and support resources (Help, Contact, Privacy) in separate white card boundaries. This logical sequencing allows users to quickly locate desired functions without cognitive overload. The Delete Account action is visually distinguished with red styling, following the principle that important or destructive actions should be clearly differentiated from standard menu options.

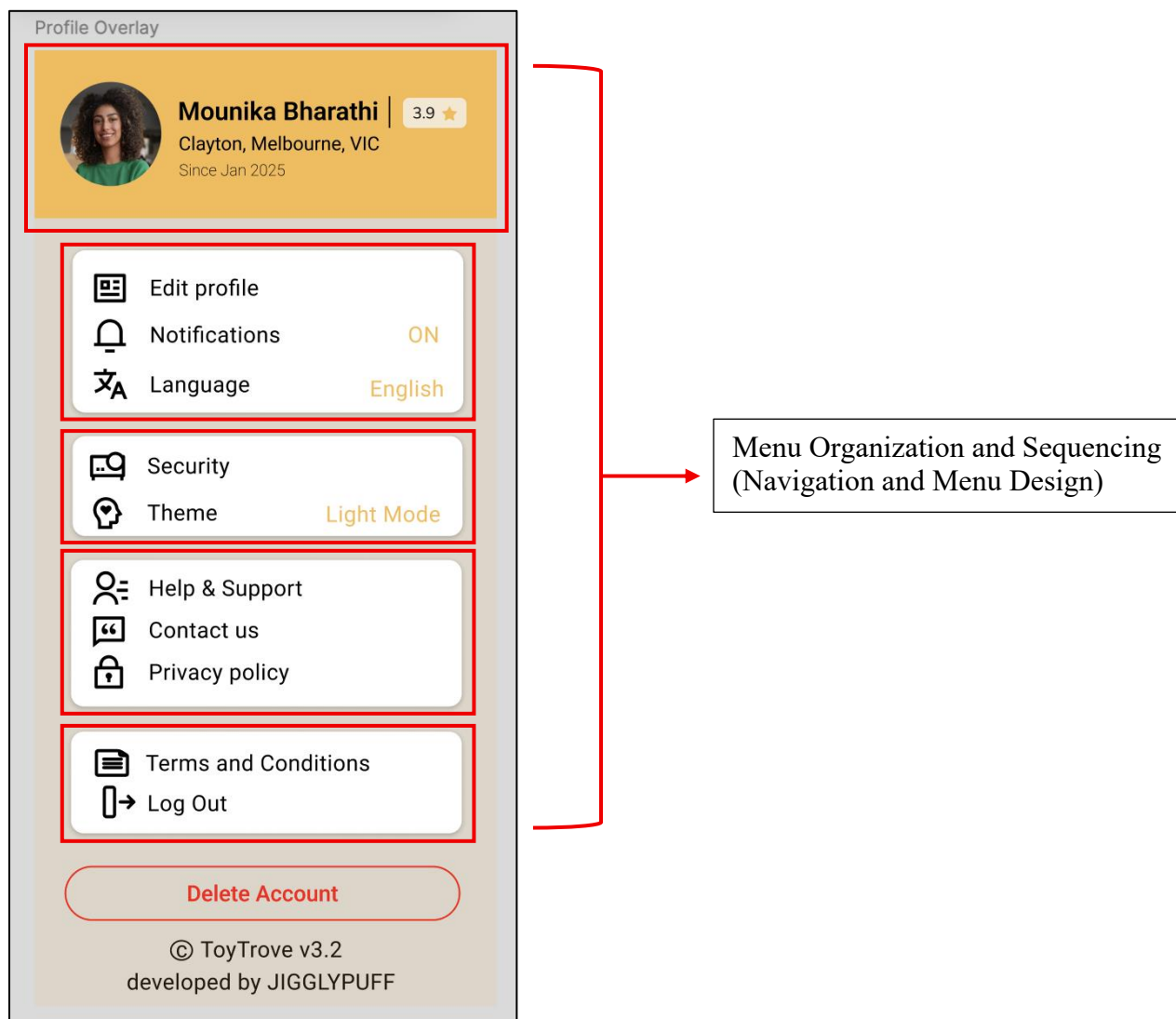


Figure 8: Profile overlay

Screen 7 – History

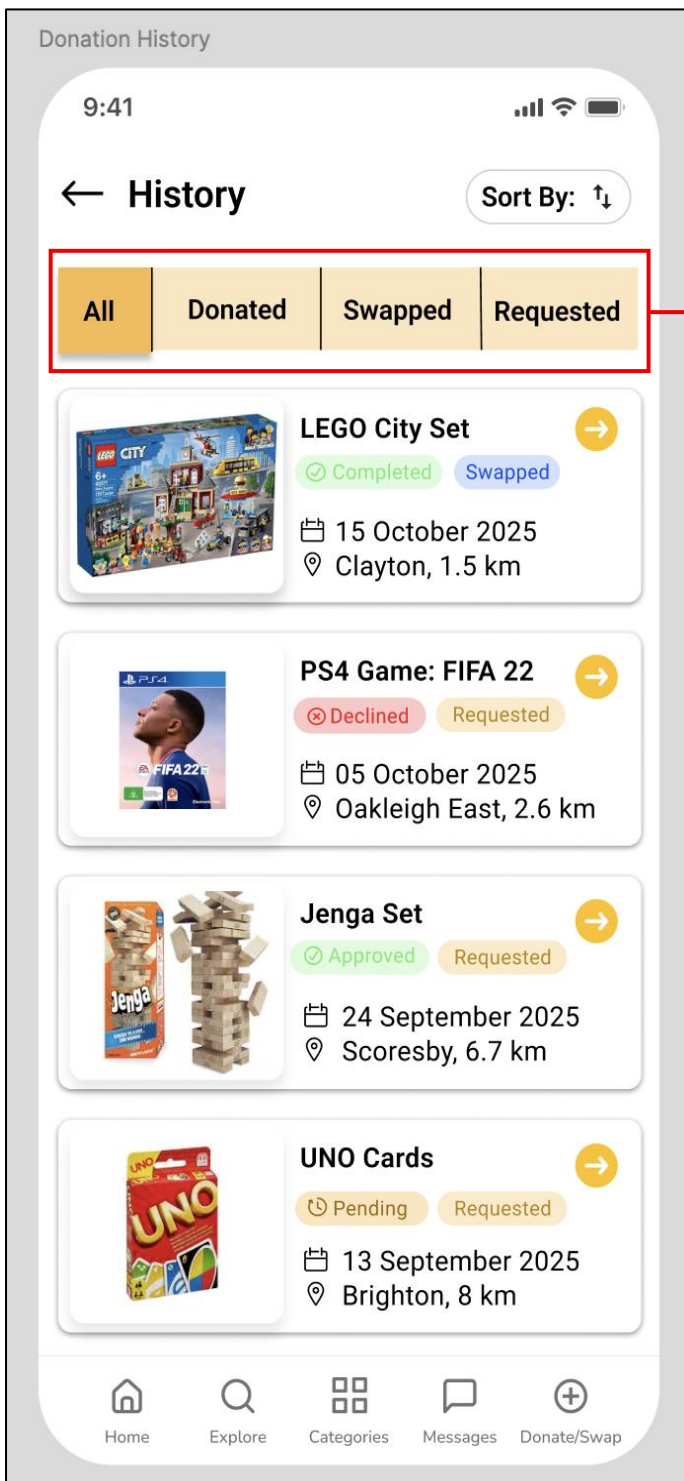
User Requirement:

Users should be able to access and manage a personal dashboard that tracks all toys they have shared, swapped, or received.

Design Principle:

Fixed Tabs Navigation (Navigation and Menu Design):

This history screen effectively implements **Fixed Tabs** for content filtering and organization. According to navigation design guidelines (Tidwell et al., 2020), fixed tabs are used for displaying important categories where all items in the set are equally important. The screen uses four equally-wide tabs (All, Donated, Swapped, Requested) with consistent text-only labelling, following the principle that tab width should remain the same regardless of label width. This design suits the 3-5 items guideline for fixed tabs, allowing users to quickly switch between transaction categories without scrolling. The tab content remains distinct, enabling efficient filtering of history items based on transaction type.



Fixed Tabs Navigation
(Navigation and Menu Design)

Figure 9: History

Screen 8 – List Item for Donation/Swap

User Requirement:

Users should be able to contribute toys in bulk with clear details about quantity, condition, and age suitability.

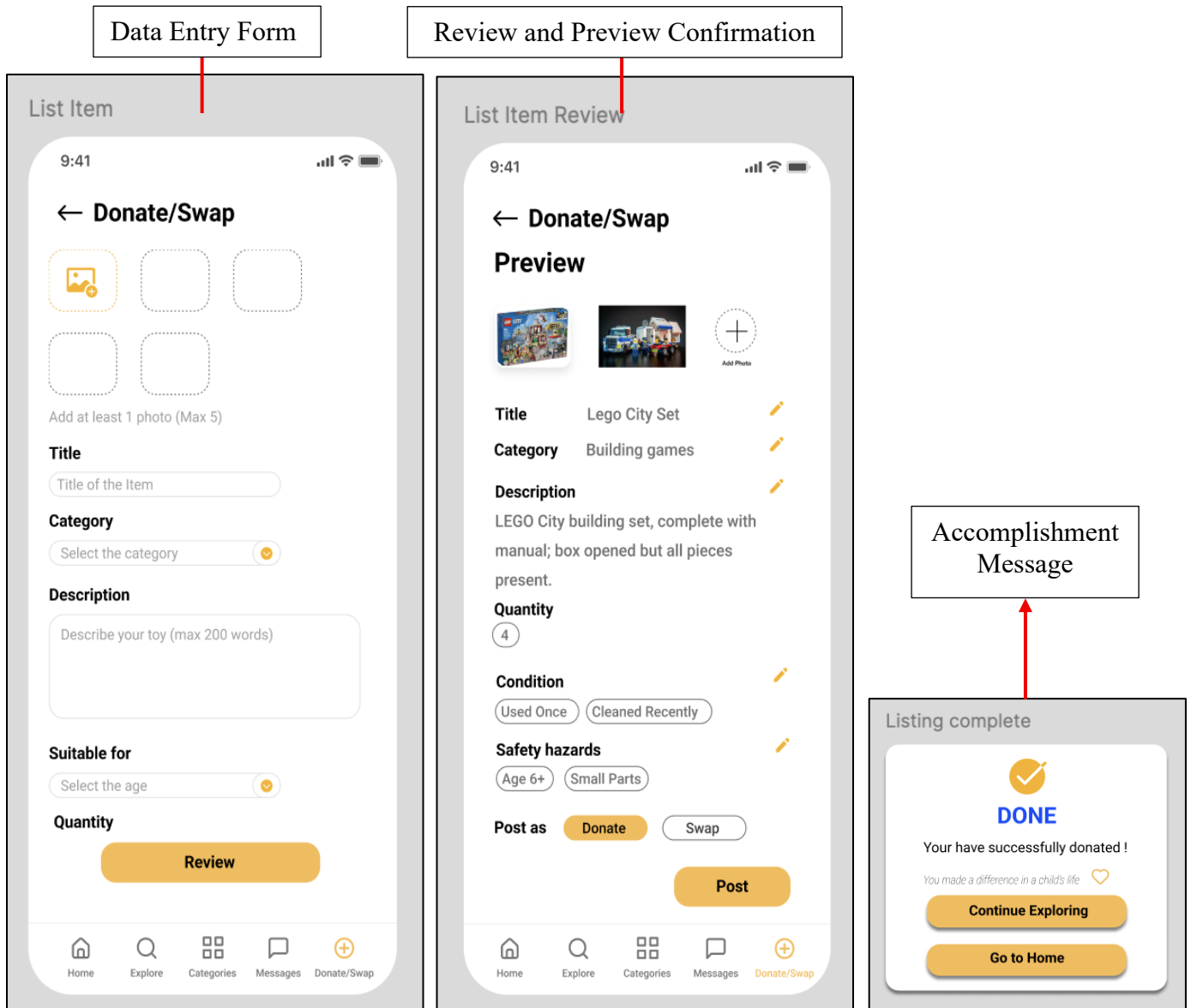


Figure 10: List item, List item review and Confirmation message

Design Principle:

Design Dialogs to Yield Closure (Shneiderman's Golden Rule - 4)

This listing form implements **Shneiderman's Golden Rule 4: Design Dialogs to Yield Closure** (Shneiderman et al., 2016). The donation process structures actions with clear beginning (data entry form), middle (review and preview confirmation), and end (success screen with accomplishment message). According to Shneiderman et al. (2016), completion feedback provides users satisfaction of accomplishment and relief. After posting, the success screen displays a checkmark, "DONE" message, and emotional reinforcement ("You made a difference in a child's life"), explicitly signaling task closure. The provision of next actions ("Continue Exploring" or "Go to Home") maintains the interaction cycle, preventing the dialogue from feeling abandoned. This complete sequence ensures users feel confident about their bulk toy contribution with clear accuracy verification and satisfying completion.

3 Accessibility Guidelines: Perceivable, Operable, and Understandable

1. Use of Colour (Level A)

Principle: 1. Perceivable

■ Guideline: 1.4 Distinguishable

Success Criterion: 1.4.1 Use of Colour (Level A) (How to Meet WCAG, 2025)

Colour is not used as the only visual means of conveying information, indicating an action, prompting a response, or distinguishing a visual element.

The prototype implements the principle that colour is not used as the only visual means of conveying information. On the Listing Error screen, the required "Title" field uses both red text ("Title of the Item is required!") and a red border outline, ensuring users with colour blindness can identify the error through multiple visual cues. The Browse and History screens display toy condition using coloured badges ("Like New" in green, "Gently Used" in yellow, "Noticeable Wear" in orange) paired with descriptive text labels, ensuring the condition information is accessible without relying solely on colour perception. Transaction statuses ("Completed", "Declined", "Approved", "Pending") combine colour-coded badges with explicit text descriptions, allowing users with colour vision deficiencies to understand status through text rather than colour alone.

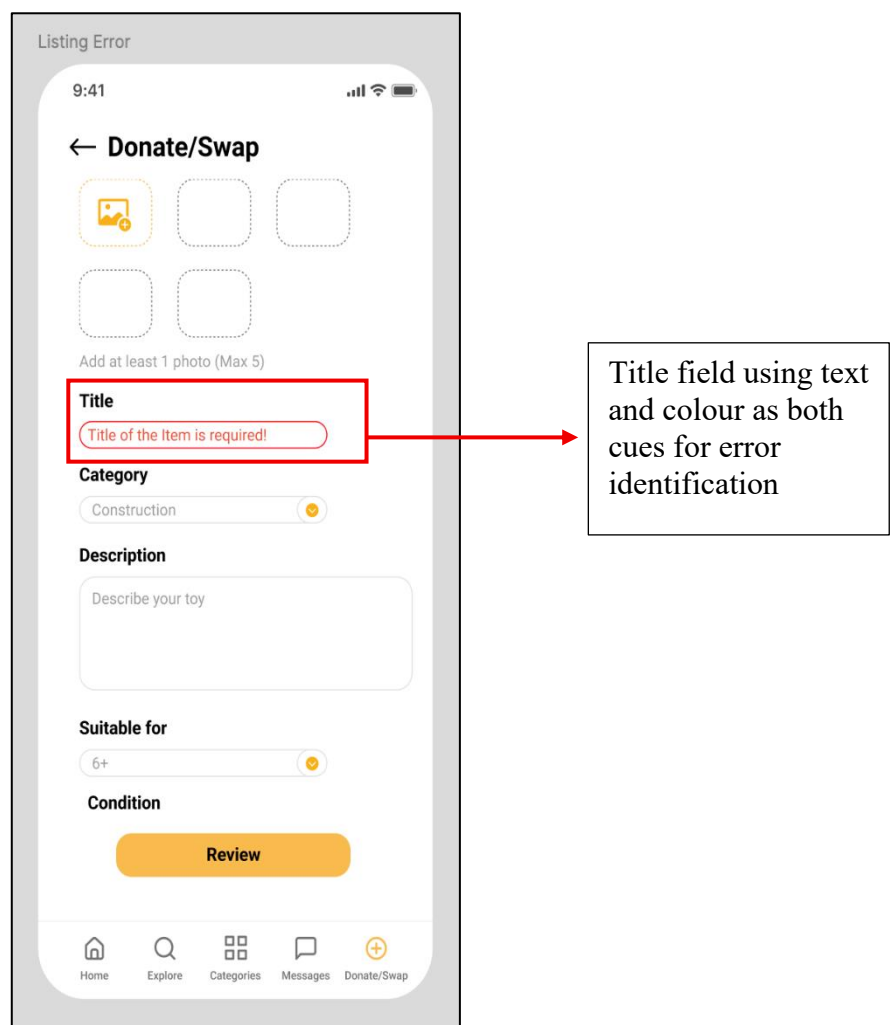


Figure 11(a): Title description - colour+text cues

Condition tags are coloured badges with descriptive labels

Transaction statuses combine colours with text description

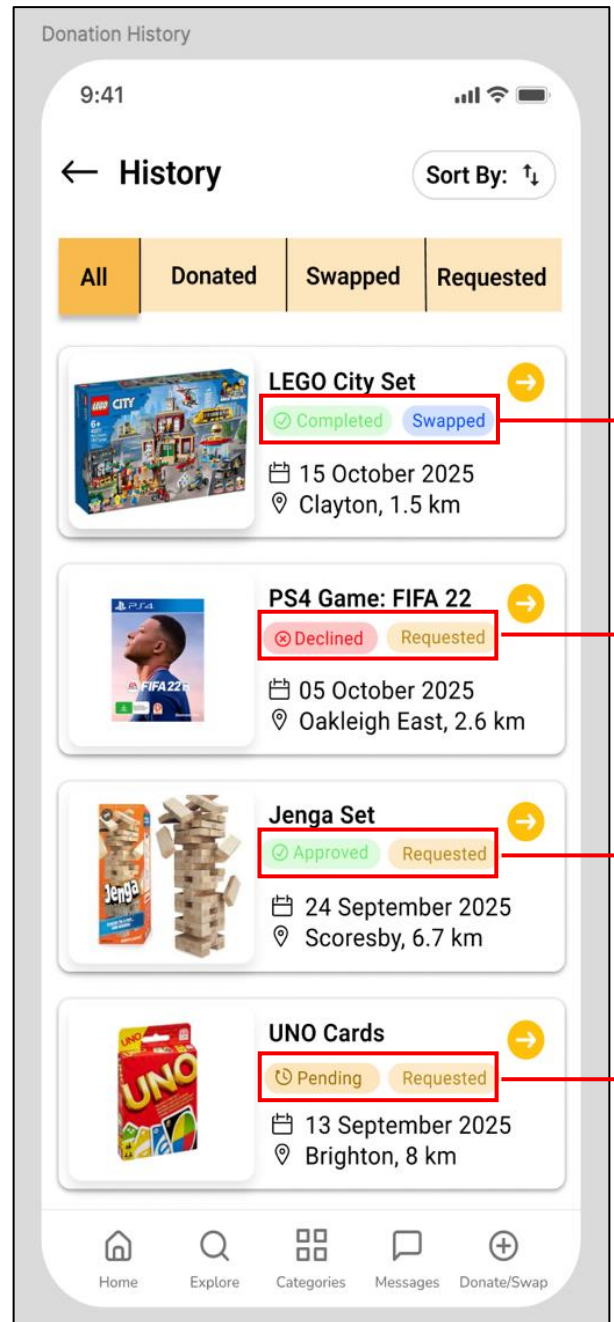
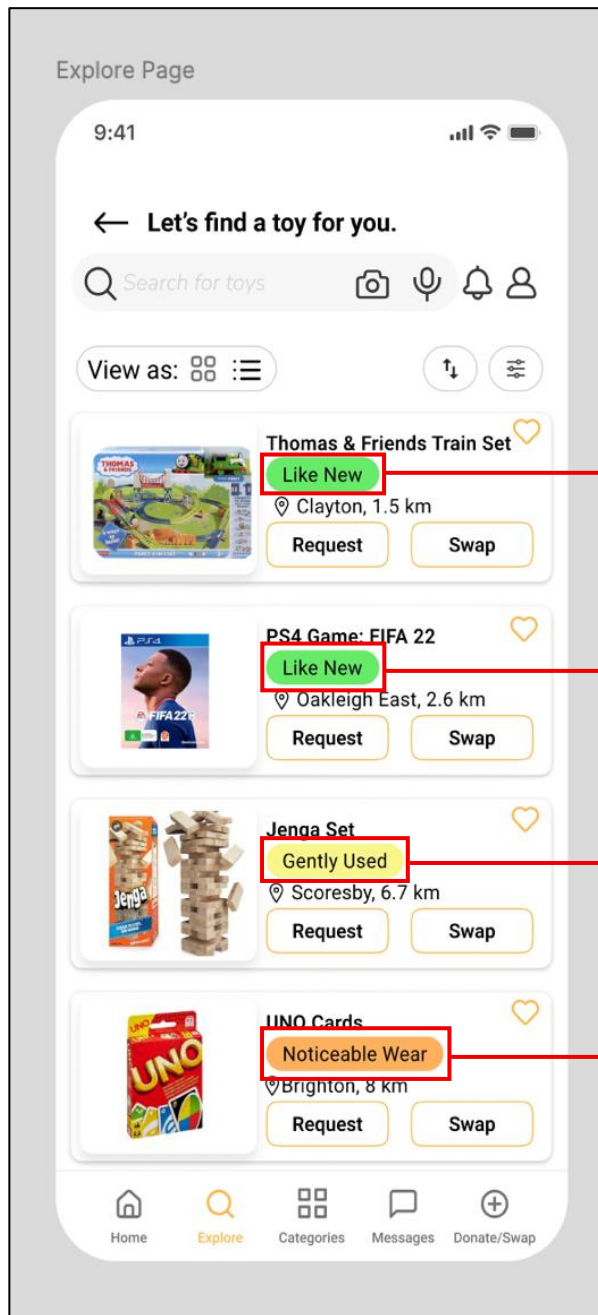


Figure 11(b): Product description(left) & Transaction status(right)
with coloured badges and text description

2. Concurrent Input Mechanisms (Level AAA)

Principle: 2. Operable

■ Guideline: 2.5 Input Modalities

Success Criterion: 2.5.6 Concurrent Input Mechanisms (Level AAA) (How to Meet WCAG, 2025)

Web content does not restrict use of input modalities available on a platform except where the restriction is essential, required to ensure the security of the content, or required to respect user settings.

Implementation in search functionality:

The search interface implements concurrent input mechanisms by supporting multiple simultaneous input methods without restriction. Users can search for toys through traditional text input in the search bar, voice commands via the microphone icon which activates "Troy" voice assistant, or visual search using the camera icon for image recognition and barcode scanning. The camera feature displays a viewfinder allowing users to photograph toys for instant product matching, while the voice assistant provides conversational search with suggested quick actions. This multimodal design benefits users with motor disabilities who prefer voice input, users with speech impairments relying on text or visual methods, and users with cognitive challenges who find visual matching easier than typing product names.

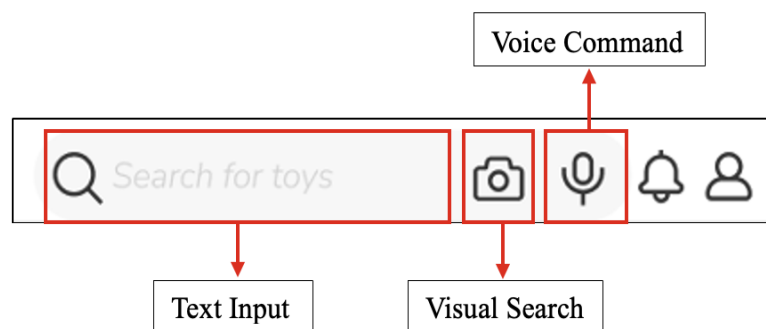


Figure 12(a): Multiple input mechanisms

Camera Search:

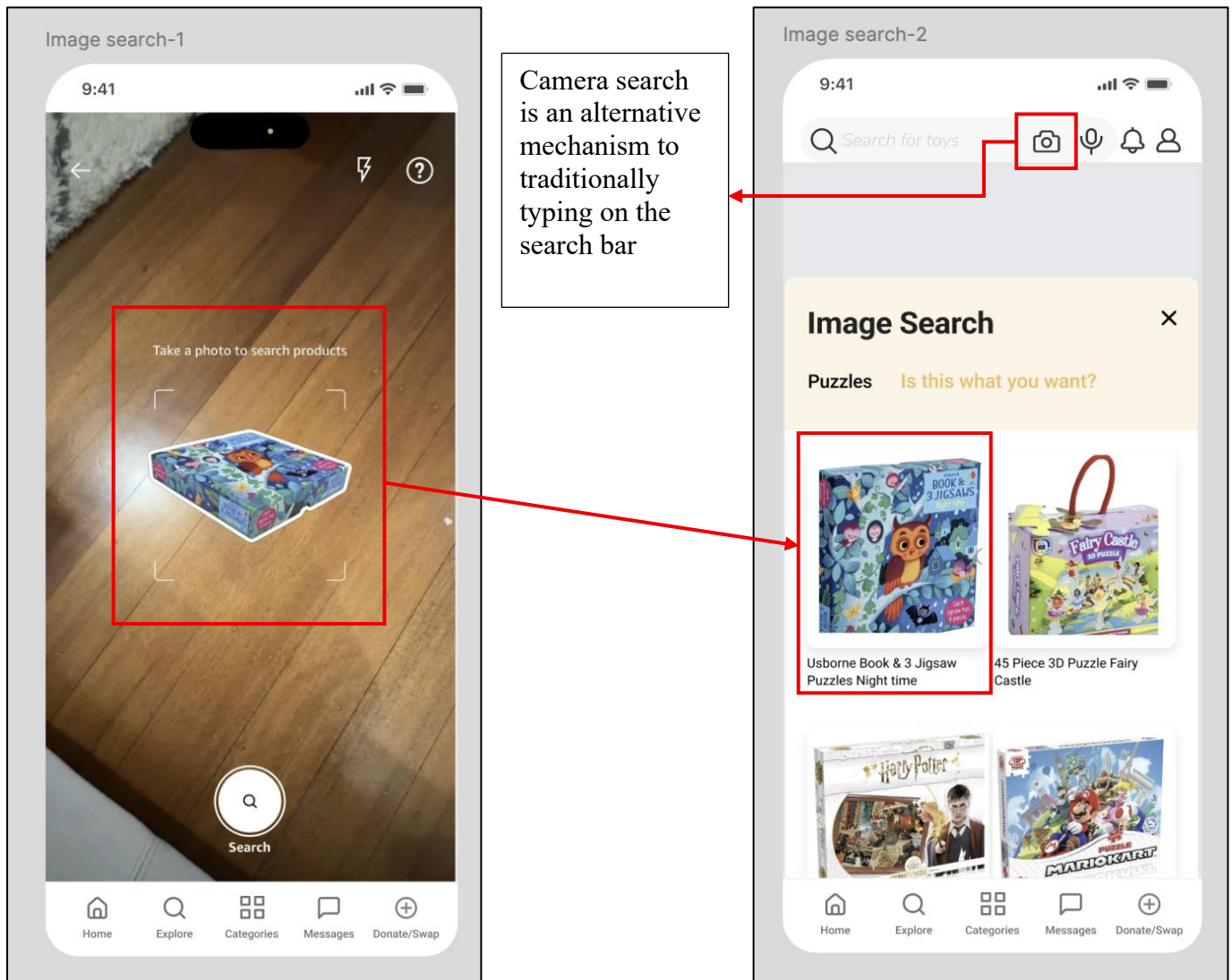


Figure 12(b): Camera search yields similar product.

Voice Search:

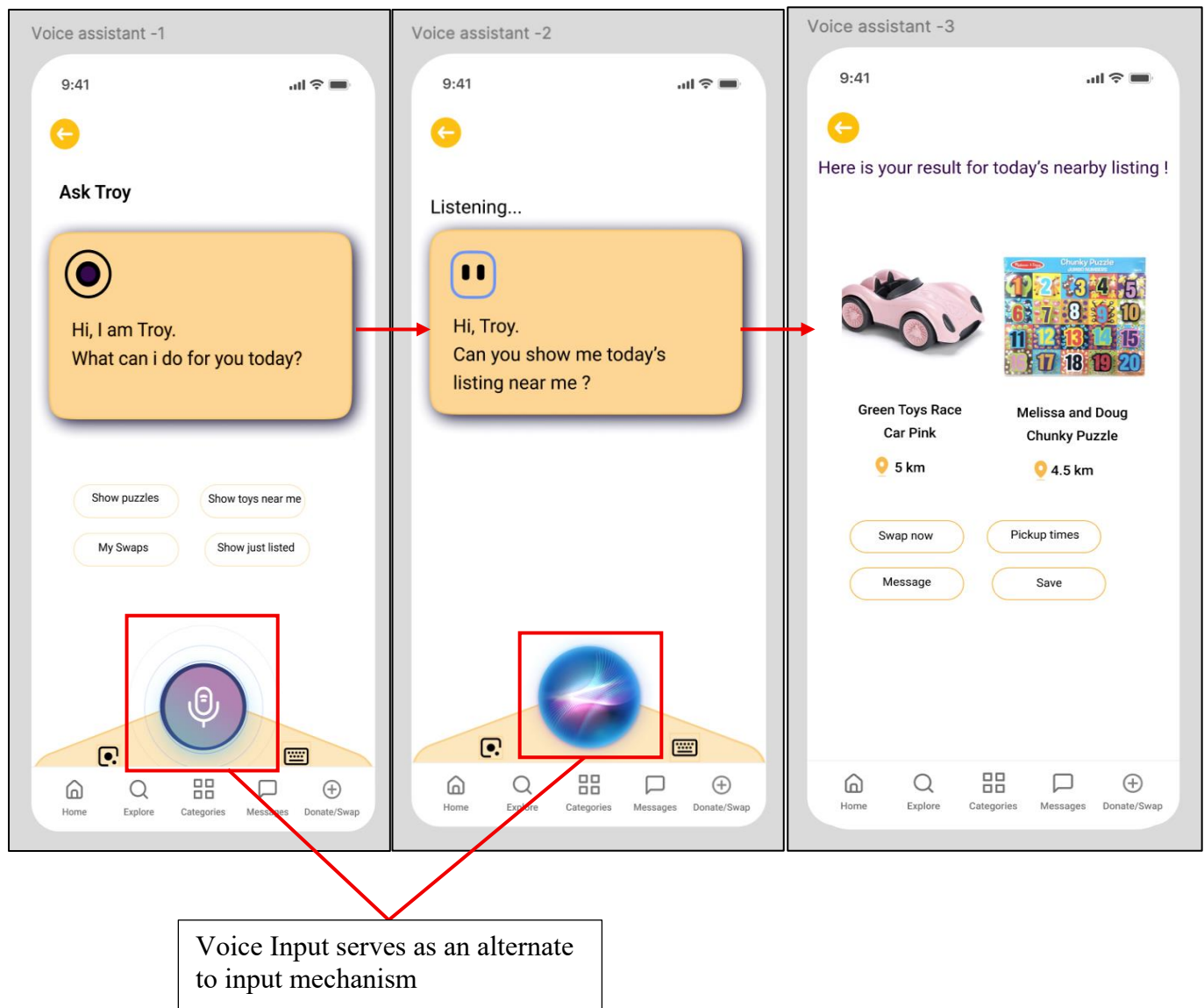


Figure 12(c): Voice search helps in searching the toys and reducing the time taken to type.

3. Error Prevention (Legal, Financial, Data) (Level AA)

Principle: 3. Understandable

■ Guideline: 3.3 Input Assistance

Success Criterion: 3.3.4 Error Prevention (Legal, Financial, Data) (Level AA)

(How to Meet WCAG, 2025)

For web pages that cause legal commitments or financial transactions for the user to occur, that modify or delete user-controllable data in data storage systems, or that submit user test responses, at least one of the following is true:

Reversible

Submissions are reversible.

Checked

Data entered by the user is checked for input errors and the user is provided an opportunity to correct them.

Confirmed

A mechanism is available for reviewing, confirming, and correcting information before finalizing the submission.

The listing form implements error prevention for user-controllable data through a multi-step review mechanism before finalizing submissions. After users complete the initial form with toy details, photos, and specifications, the interface provides a dedicated Preview screen allowing users to review all entered information including title, category, description, condition, and safety hazards. Users can edit any field using the pencil icons before final submission. This confirmation step prevents accidental data loss or incorrect listings, particularly important when users donate or swap toys involving commitments. The completion screen provides clear closure with success feedback, ensuring users understand their submission is finalized, meeting the "Confirmed" requirement for data modification transactions.

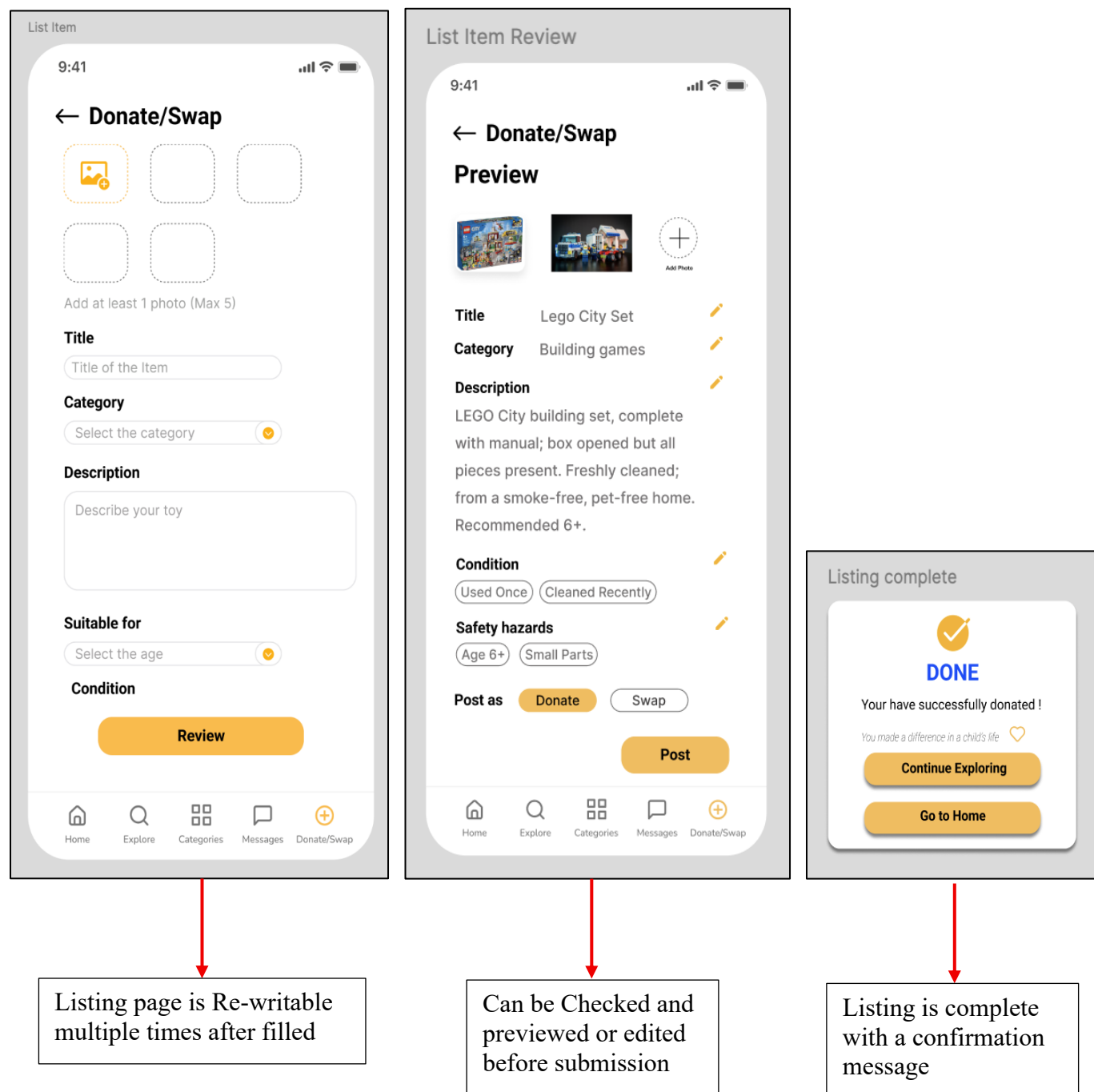


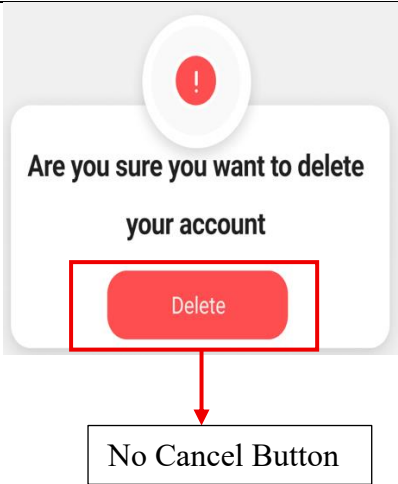
Figure 13: Listing page can be edited, reviewed, and confirmed

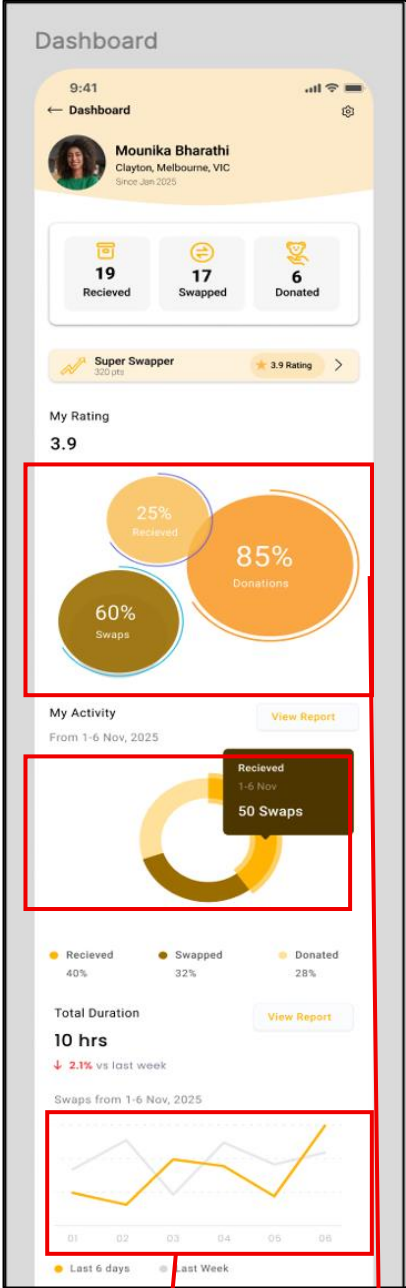
Part 3: Individual Heuristic Evaluation

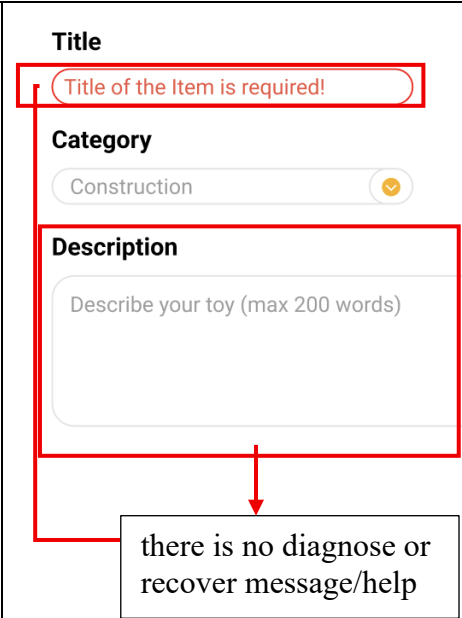
1. Student Name: Rukma Kuppusamy

Student ID:34517138

Heuristics Violations:

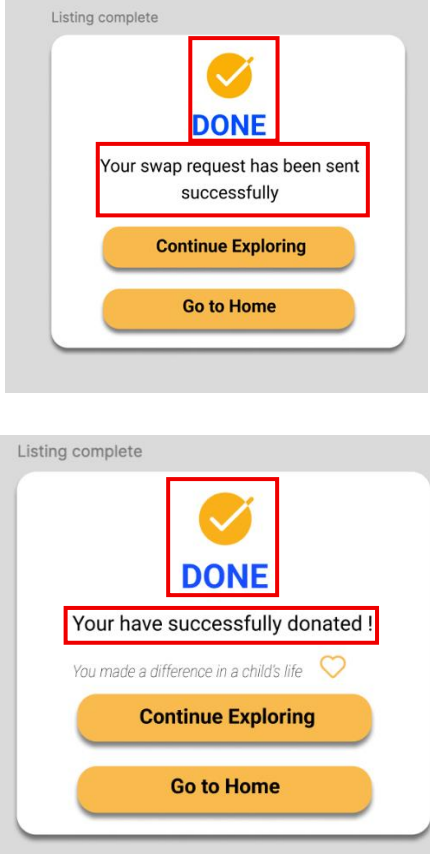
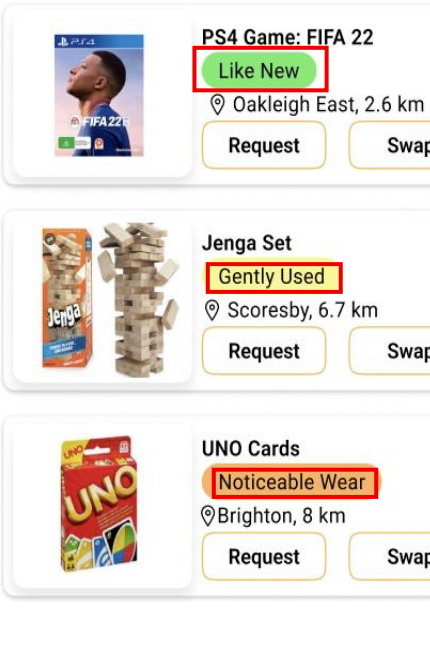
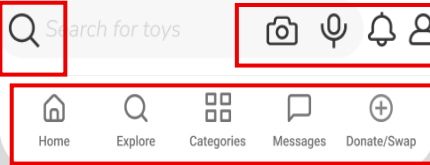
Heuristic Rule	Instance of Violation	Evidence	Severity Rating and Justification	Recommendation
H3. User control and freedom	Delete confirmation dialog shows only "Delete" button without visible "Cancel" option, trapping users in destructive action with no escape route.		Severity: 4/4 (Catastrophic) Frequency: 1 (Rare) - Users rarely delete accounts; typically once ever if at all. Impact: 4 (Critical) - Violates Shneiderman's Golden Rule 6 (Permit Easy Reversal). Creates false constraint (Norman) forcing irreversible catastrophic action. Permanent loss of entire account, all data, listings, connections, history. Creates extreme anxiety. No recovery possible.	Add standard two-button dialog: "Cancel" (left, large, primary green/neutral) and "Delete Account" (right, secondary red outlined). Add additional safety: require typing "DELETE" or account email to confirm. Show warning text: "This permanently deletes your account and all data. This cannot be undone." Theory Justification: Applies Norman's Signifiers (Cancel signals safe exit) and follows natural mapping. For catastrophic actions, Shneiderman's Rule 6 requires even stronger reversal mechanisms - multi-step confirmation prevents accidental irreversible loss.

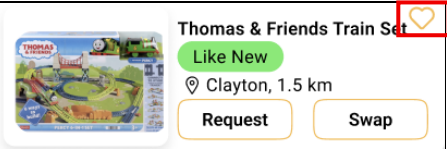
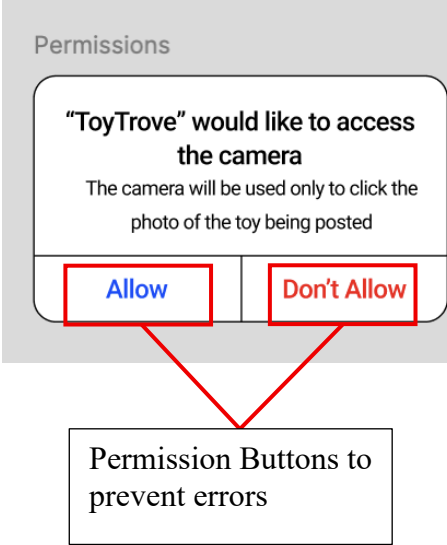
			Persistence: 1 (once)- Encountered only when attempting account deletion.	
H8. Aesthetic and minimalist design	Dashboard displays multiple complex visualizations simultaneously (graphs, charts, statistics), overwhelming users with competing information that dilutes focus from essential content.	 Too many graphs	Severity: 2/4 (Minor) Frequency: 3 (Moderate-High) - Seen every session as primary dashboard screen. Impact: 2 (Moderate) - Violates Miller's Law (exceeds 7±2 capacity) and Gestalt Figure-Ground. Affects comprehension speed but not core functionality. Persistence: 2 (Moderate) - Repeated but users learn to ignore non-essential elements over time.	Implement progressive disclosure: display 2-3 key metrics prominently, use card-based layouts, move detailed analytics to expandable sections. We must learn to Communicate not decorate. Theory Justification: Applies Gestalt Common Region and reduces cognitive load in attention/perception processes.

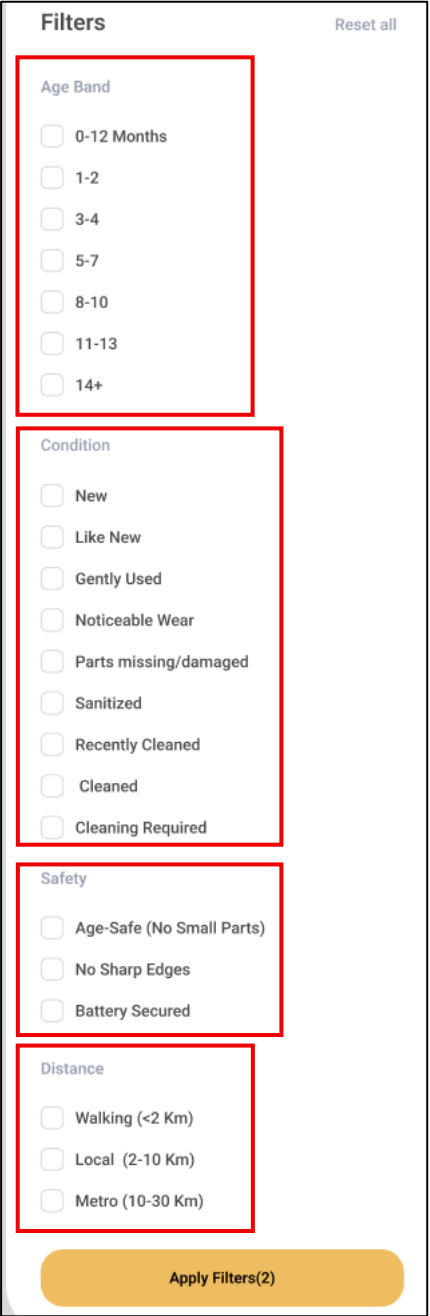
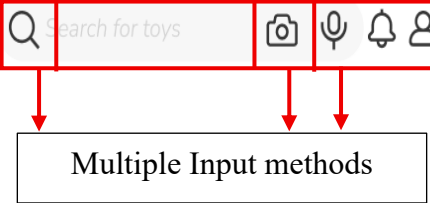
H9. Help users recognize, diagnose and recover from errors	The title input field displays the error message "Title of the Item is required!" when left empty but fails to provide constructive guidance on how to resolve the error or what constitutes a valid title entry.	 The screenshot shows a form with three sections: 'Title', 'Category', and 'Description'. The 'Title' section has a red border and a red error message 'Title of the Item is required!'. The 'Category' section has a dropdown menu with 'Construction' selected. The 'Description' section has a text area with the placeholder 'Describe your toy (max 200 words)'. A red arrow points from the 'Title' section to a box containing the text 'there is no diagnose or recover message/help'.	Severity: 2 Frequency: 3/4 - Users commonly submit forms with missing fields, especially first-time users. Impact: 2/4 - Users understand the problem exists but lack guidance on how to fix it, causing minor frustration without blocking completion. Persistence: 2/4 - Must overcome each time the error occurs until users learn what constitutes an acceptable title.	Revise the error message to "Please enter a title for your item (e.g., 'Wooden Building Blocks')" to provide constructive recovery guidance. This applies Shneiderman's Golden Rule #5 (Prevent Errors) by offering specific instructions, and uses Norman's Principle of Feedback to ensure informative, actionable error messages that help users recover successfully.
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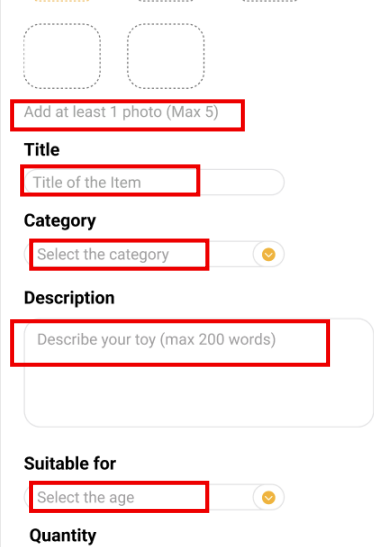
Heuristics Compliances:

Heuristic Rule	Instance of Compliance	Evidence	Design Justification
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H1: Visibility of System Status	The tick checkmark icon appears prominently at the top centre of both completion screens, immediately above the blue "DONE" text, accompanied by explicit text confirmation messages: "Your swap request has been sent successfully" and "You have successfully donated!" that clearly communicate task completion status.		This compliance applies Norman's principle of Feedback, which states that feedback provides status update information about interactions that are in progress or have been completed. The interface provides visible, clear, and timely feedback about task completion, fulfilling Shneiderman's Golden Rule #3 (Offer Informative Feedback). The feedback is tightly coupled to the relevant interaction, clearly indicating that the task is completed, which aligns with best practices for visibility of system status.
H2. Match between system and the real world	Condition indicators use natural language: "Like New," "Gently Used," "Noticeable Wear" instead of technical grades (A/B/C or numerical scores).		Speaks users' language with familiar real-world concepts rather than system-oriented terms. Aligns with Norman's Conceptual Models by leveraging existing mental models from real toy-sharing contexts. Reduces cognitive load - zero learning curve, instant recognition, no memorization needed. Applies knowledge in the world vs. knowledge in the head principle.
H4. Consistency and standards	Interface consistently uses industry-standard icons: camera (photo upload), heart (favourites),		Follows Jakob's Law - users prefer interfaces working like familiar sites. Icons function as Norman's Signifiers creating perceived

	filter (search refinement), profile (account access), Home button, magnifying lens(explore),plus(add listing),chat bubble(messages).		affordances. Reduces cognitive load through transfer learning from other platforms. Supports Gestalt Similarity principle - consistent styling signals "selectable options across contexts."
H5. Error prevention	The permissions dialog displays a confirmation prompt with the heading "ToyTrove would like to access the camera" followed by explanatory text "The camera will be used only to click the photo of the toy being posted", requiring explicit user consent before granting camera access through two clearly labelled options: "Allow" and "Don't Allow".		This compliance implements Norman's Principle of Constraints , which limits available actions to prevent users from accidentally entering invalid states or granting unintended permissions. The design follows Shneiderman's Golden Rule #5 (Prevent Errors) by requiring explicit confirmation before accessing sensitive device features. The explanatory text provides transparency about the permission's purpose, helping users make informed decisions and preventing mistakes. The clear binary choice uses psychological constraints to ensure deliberate user action, reducing accidental permission grants.

H6. Recognition rather than recall	Filter screen displays all options in four categories (Age, Condition, Safety, Distance) with checkboxes, eliminating need to remember available filters.		Minimizes memory load by making options visible rather than requiring recall. Directly implements Shneiderman's Golden Rule 8 (Reduce Short-Term Memory Load) - avoids interfaces where users must remember information from one display to use on another.. Applies Gestalt Common Region - each category in distinct bounded region creates clear grouping. Transforms memory-intensive task into simple visual recognition and selection.
H7. Flexibility and efficiency of use	Description input offers three methods: text typing (novice), camera icon (intermediate), voice icon (expert accelerator for fastest entry).		Provides accelerators for experts while maintaining novice accessibility. Supports Shneiderman's Golden Rule 2 (Seek Universal Usability) accommodating different skills, abilities, and contexts. Each method provides Norman's Signifiers indicating specific affordances. Voice input offers genuine efficiency gains (2-3x faster than typing)

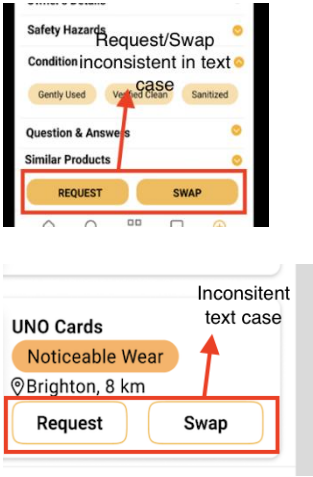
			following progressive complexity principles.
H10. Help and documentation (Proactive)	The listing form provides inline instructional text and placeholder guidance in multiple input fields: "Add at least 1 photo (Max 5)" specifies photo requirements, "Describe your toy (max 200 words)" indicates character limits, and dropdown placeholders "Select the category" and "Select the age" provide clear entry instructions directly within the form fields.		This compliance implements proactive help through contextual guidance directly within the interface. The inline instructions provide help at the point of use, eliminating the need for users to seek external documentation. This aligns with Shneiderman's Golden Rule #8 (Reduce Short-Term Memory Load) by displaying requirements and constraints like "(max 200 words)" and "(Max 5)" at the moment of need, making the system easier to use while guiding users toward successful task completion.

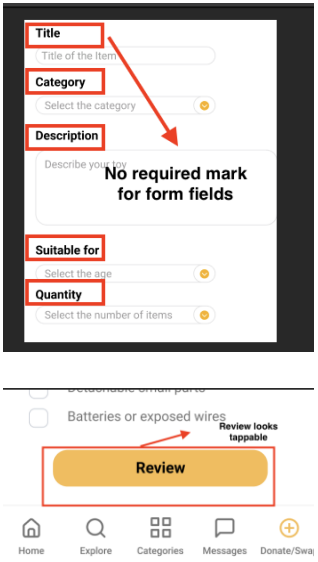
2. Student Name: Rachit Sharma

Student ID: 35203528

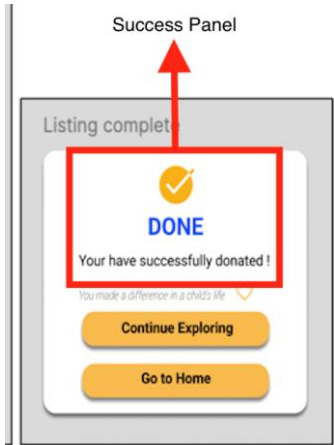
Heuristics Violations:

Heuristic Rule	Instance of Violation	Evidence	Severity Rating and Justification	Recommendation
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H4 Consistency & standards	Primary CTAs are inconsistent: “Request/Swap” (title case on Explore cards) vs “REQUEST/SWAP” (all caps on Product page). Users can’t rely on a single pattern.		Severity 2 (Minor) — Frequency: on multiple core screens (common). Impact: slows decisions / increases hesitation. Persistence: persists whenever moving between list and detail. Breaks H4’s guidance to use familiar, standard words and styles	Create one button token: label = “Request swap” (Title Case), same size/weight/color everywhere. This applies Jakob’s Law (familiarity reduces learning time) and menu best-practice (consistent patterns).
H6 Recognition rather than recall	The Sort control doesn’t reveal the currently active sort option. The trigger button displays generic placeholder text (“Sort by…”) regardless of selection, and the dropdown menu lacks visual indicators (checkmark, highlight, or bold text) showing which option is active. Users must remember their previous selection.		Severity 2 (Minor) — Frequency: any time users sort. Impact: hesitation/mis-sort. Persistence: repeats whenever refining. Forces short-term recall over recognition (H6); increases working-memory load (Miller’s Law).	Reflect the active choice on the trigger (e.g., “Sort: Newest”) and ✓ the selected option in the menu; keep the selection persistent across sessions. This externalises state (H6), reduces memory burden (Miller), and speeds selection (Hick’s Law via fewer mental steps).

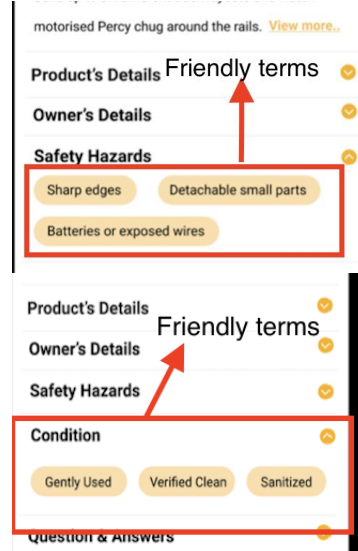
H9 Help users recognise, diagnose & recover from errors	List Item form omits clear “required” indicators; Review appears tappable while required fields are incomplete.		Severity 3 (Major) — Frequency: every new listing. Impact: users repeatedly try “Review” without knowing what’s missing (trial-and-error, high friction). Persistence: recurs for each required field until corrected. H9 expects clear language about what’s wrong and how to fix it.	Make “Review” truly disabled until all mandatory fields validate ; Add visible ‘required’ indicators to mandatory fields. This prevents invalid actions (Norman’s constraints & immediate feedback), lowers working-memory load (Miller) because users don’t have to remember rules, and matches familiar mobile patterns (Jakob) for first-time success.
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Heuristics Compliances:

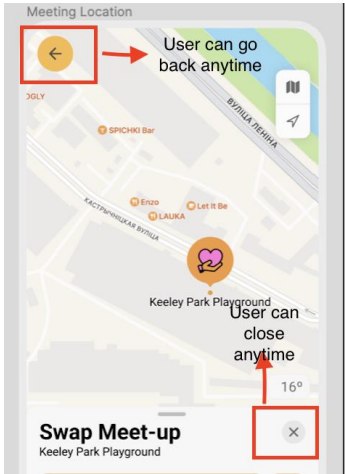
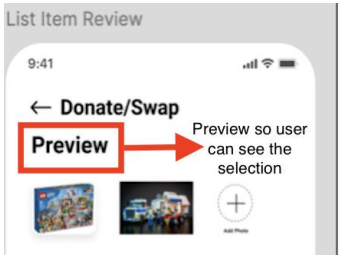
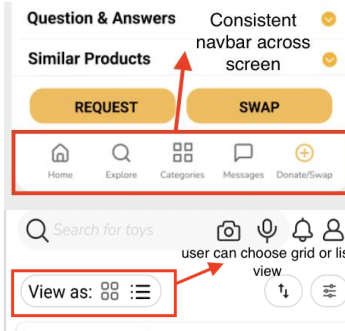
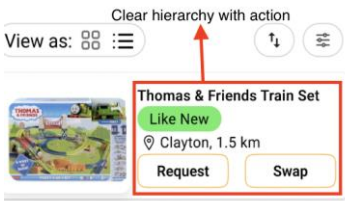
Heuristic Rule	Instance of Compliance	Evidence	Design Justification
H1 Visibility of system status	Listing complete card: “DONE — You have successfully donated” + next actions.		Concise, prompt feedback completes the user's goal loop and mitigates redundant taps. This fits with WCAG 3 – Understandable (expected outcomes, assistance in error avoidance) and WCAG 4 – Robust (state changes must be programmatically communicated for assistive technology), hence enhancing trust for all users, including those utilising assistive technology. Users benefit from faster goal closure and fewer re-checks; for screen readers, the status is announced.

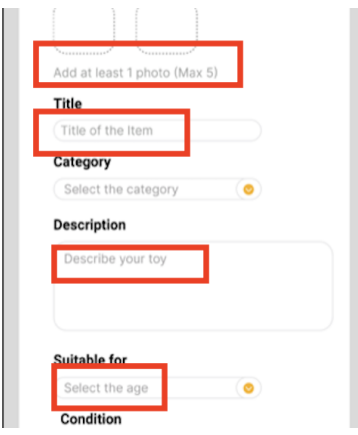
H2 Match between system & real world

Condition & Safety tags use plain, parent-friendly terms (e.g., *Gently used*, *Detachable small parts*).



Familiar, parent-friendly terminology diminishes interpretation time (reducing cognitive load for busy parents multitasking) and prevents misunderstandings (e.g., parent immediately knows 'Detachable small parts' means choking hazard without having to decode technical safety terminology). In addition to H2, this aligns with WCAG 3 – Understandable (“Text is readable and comprehensible; content is presented in predictable formats”), enhancing skim comprehension and diminishing cognitive burden. Users benefit from reduced interpretation time and fewer misreads, which aligns with Plain Language and WCAG understandable patterns.

H3 User control & freedom	Meeting Location has Back and Close (×), so users can exit safely.		Users may abandon a path without incurring any penalties (undo/escape). This reflects the principles of Structural Navigation, where uniform controls enable users to swiftly reset or modify goals, so aligning with WCAG 2 – Operable (facilitating ease of navigation and location identification). Users benefit from reduced mode errors and sunk-cost lock-in, which supports their ability to make ‘graceful exits’.
H5 Error prevention	Preview step before posting		Guardrails inhibit invalid submissions at the outset (prevention supersedes remedy). This closely aligns with WCAG 3 – Understandable (aiding users in avoiding and rectifying errors) and minimises rework, particularly on mobile platforms where correction expenses are substantial.
H7 Flexibility & efficiency of use	View as (grid/list) + persistent bottom tab bar.		Experts act faster via shortcuts while novices retain clarity. This applies Jakob’s Law (familiar patterns lower learning time) and Miller’s Law (toggling surfaces just the needed representation, cutting working-memory load). Persistent global nav also follows structural-nav best practice (reset/change goals quickly).
H8 Aesthetic & minimalist design	Product cards show clear hierarchy (title › condition › actions) with space and limited colour.		Good visual hierarchy and restrained colour direct attention and improve discoverability; too many colours overwhelm, so the minimal set here supports faster scanning. Fewer competing hues reduce the number of perceived choices, lowering decision time per Hick’s Law, and reduce working-memory load per Miller’s Law. Using neutral containers with accent colour only for primary cues also aligns with WCAG guidance

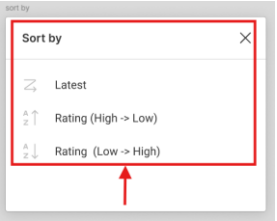
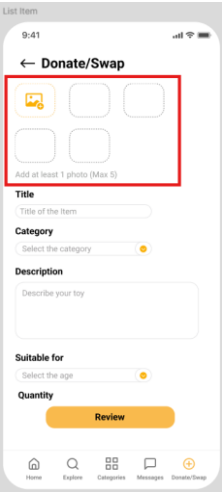
H10 Help & documentation	Inline info icons beside form fields give at-point-of-need help.		Proactive, in-context help reduces ambiguity at the moment of need. This matches Nielsen H10's guidance on proactive vs reactive help and supports WCAG 3 – Understandable (predictable controls; help to avoid mistakes). Net effect: fewer stalls, smoother first-time success.
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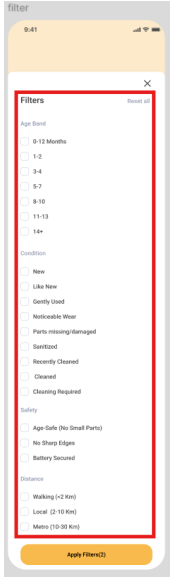
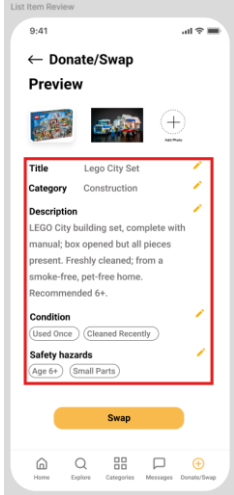
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Student ID: 34525416

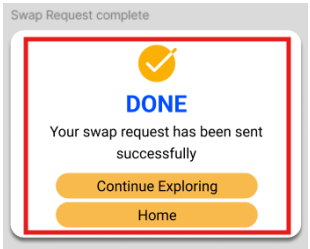
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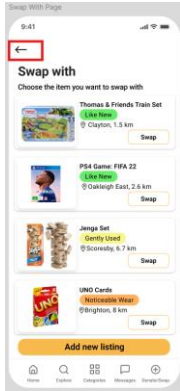
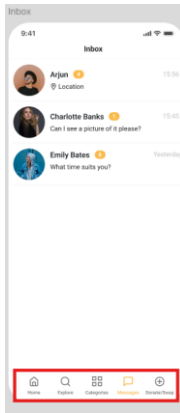
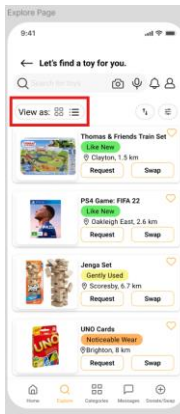
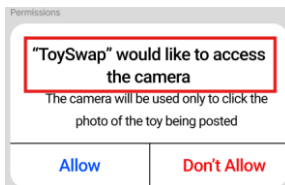
Heuristic Rule	Instance of Violation	Evidence	Severity Rating and Justification	Recommendation
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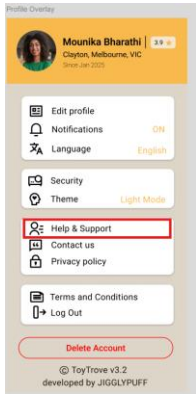
H2: Match Between System and Real World	The "Sort by" overlay uses terminology that doesn't match how users think about toy browsing. Terms like "Rating (High > Low)" and "Rating (Low > High)" are unclear - what does "rating" actually mean here? The ">" symbol feels like coding language rather than everyday speech. When I think about how parents actually browse for toys, they're looking for "most popular," "most requested," or "recently posted" items.		Severity: 3 Frequency: High - This will affect everyone who tries to sort their search results, so it's quite common. Impact: Moderate - confuses sorting criteria, may lead users to avoid the sorting feature entirely Persistence: Repeated - occurs whenever users want to organise their search.	Replace technical terminology with user-friendly language: "Most Popular First," "Most Requested First," "Recently Added," "Nearest to Me." Remove technical symbols (>) and use plain language. Consider using icons alongside labels (e.g., a star for quality, a clock for recency).
H5: Error Prevention	Users can reach the "Review" stage without uploading photos. This is problematic, since our survey showed that 57% worry about hygiene and safety—photos are essential for trust—but the system allows photoless listings that undermine platform confidence.		Severity: 3 Frequency: Moderate - occurs when users skip photo upload Impact: High - directly conflicts with core safety/trust requirement Persistence: Must be resolved each time before posting	Disable the "Review" button until at least one photo is uploaded. Add helper text: "At least 1 photo required - helps others trust your listing!" Include progress indicator: "1/5 photos added."

H6: Recognition Rather Than Recall	Filter lists show no indication of which options have results. Users must remember selections and guess whether toys match their criteria before applying filters, forcing reliance on memory rather than recognition.		Severity: 2 Frequency: Moderate-affects all users using filters while browsing items Impact: Moderate - increases cognitive load, leads to empty result frustration Persistence: Repeated during each browsing session	Add counts next to options: "Like New (23)," "Gently Used (47)." Grey out zero-result filters. Show "47 items match our selections" at the top. We can add sticky filter chips showing active selections.
H8: Aesthetic and Minimalist Design	The Review screen shows everything at once - photos, editable fields, tags, description, conditions, safety hazards, and buttons. The critical "Post as Donate/Swap" decision is buried at the bottom, overwhelming users at the conversion point.		Severity: 3 Frequency: High - affects all users finalising listings Impact: Moderate - may cause posting errors or abandonment Persistence: Once per listing, but at a crucial decision moment	We can use progressive disclosure: show summary by default with expandable sections for "Item Details," "Safety Info," and "Description." Move the "Post as" toggle to the top with strong visual differentiation. Separate read-only info from editable fields using a clear hierarchy.

Heuristic Compliances:

Heuristic Rule	Instance of Compliance	Evidence	Design Justification
H1: Visibility of System Status	The "DONE" success screen shows a clear checkmark with explicit confirmation: "our swap request has been sent successfully" or "You have successfully donated!"		This implements Ben Shneiderman's Golden Rule H5 (offer informative feedback). The checkmark, confirmation text, and clear next steps ("Continue Exploring," "Go to Home") create a complete feedback loop. Critical for trust-building since

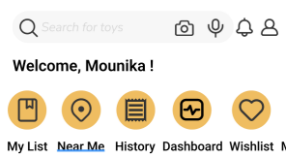
			our survey showed trust concerns as the top barrier (34.5% of responses).
H3: User Control and Freedom	Every screen has back arrow navigation (top-left), and overlays include "X" close buttons. Users can exit flows freely without forced completion.		Nielsen states users need clear "emergency exits" after mistakes. The standard back placement addresses our survey finding that 45.2% cite complicated interfaces as barriers. This freedom encourages exploration - essential for our "Hesitant Requester" persona, building platform trust.
H4: Consistency and Standards	The bottom navigation bar maintains consistent placement and iconography across all screens. The five core functions use our app's standard icons with orange/gold active state highlighting.		Nielsen's research shows consistency reduces learning time through knowledge transfer. Following iOS Human Interface Guidelines matches user mental models from familiar apps. For our tech-savvy user base (61.7% under 35), this supports our user story requiring "clear navigation" through predictable, persistent positioning.
H7: Flexibility and Efficiency of Use	The Explore page offers grid/list view toggle, plus "Request" and "Swap" action buttons on item cards for single-tap interactions.		Nielsen's heuristic emphasises supporting novices and experts through shortcuts. View switching lets detail-oriented users see more in list view, while visual browsers scan grids. Card action buttons reduce navigation depth for our "time-poor, cash-rich" segment (50% of full-time employees). Implements Shneiderman's Rule H7 (support user control).
H9: Help Users Recognise, Diagnose, and Recover from Errors	The camera permission dialogue explains why access is needed ("to click the photo of the toy being posted") with clear "Allow" or "Don't Allow" choices.		Nielsen requires plain language that explains problems and suggests solutions. Following Apple's guidelines addresses our survey (Q9), where 27.7% felt uncomfortable with permissions. The transparency builds trust for a privacy-conscious persona. Users can deny and still access other features (graceful degradation).

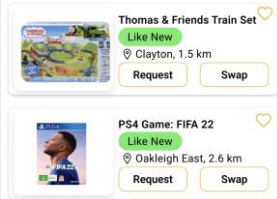
H10: Help and Documentation	The profile menu groups help resources logically: "Help & Support," "Contact us," "Privacy policy," and "Terms and Conditions" in the expected settings location.		Nielsen acknowledges that help should be easy to find when needed. This accessible structure addresses our survey finding that 45.2% struggle with "complicated settings." The prominent "Contact us" supports secondary research showing responsive customer support is critical for new user trust (Whirli/Toy Exchange reviews).
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4. Student Name: Anish Srinath

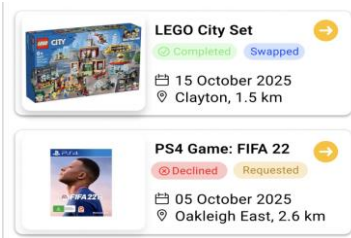
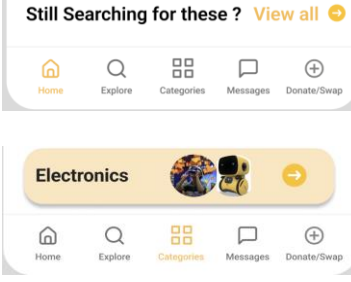
Student ID: 35292091

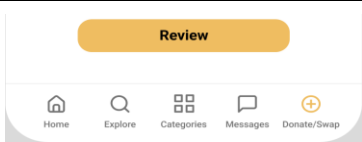
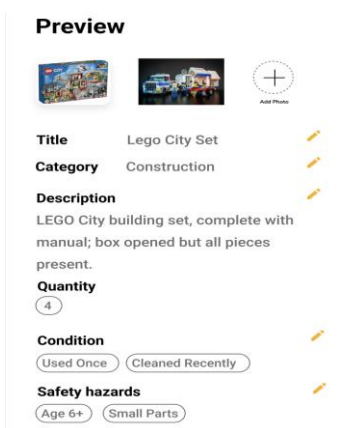
Heuristics Violations:

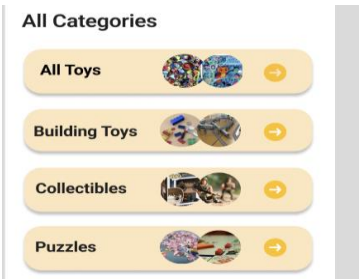
Heuristic Rule	Instance of Violation	Evidence	Severity Rating and Justification	Recommendation
Heuristic 10: Help and documentation	There is no visible help system, documentation, or tooltip to guide users through key features. New users encountering terms like "Donate vs Swap" have no way to access contextual help or explanations.		Severity: 3 - Major usability problem Frequency: This issue affects all screens consistently, making it a systemic problem. Persistence: Users will encounter this repeatedly as they navigate unfamiliar features, requiring them to use trial-and-error or external resources.	Add a clearly visible help icon (question mark) in the top navigation bar that provides contextual help. Implement tooltips for first-time users explaining key concepts like "Swap," "Donate," and status indicators.
Heuristic 3: User control and freedom	On the List Item Review page, uploaded photos cannot be edited or removed. Once users add photos, they have no way to delete them or replace them with different images if they uploaded the wrong ones or changed their mind. Users lack control over reversing this action without	Preview 	Severity: 3 - Major usability problem Frequency: Common - users frequently upload wrong photos or want to adjust their photo selection during listing creation. High impact - users must either post with incorrect photos, abandon the listing and start over, or navigate back through multiple steps, losing other entered information.	Add a delete button (X icon) to the top-right corner of each uploaded photo thumbnail. Allow users to tap the X to remove unwanted photos. Consider adding an "Edit" option that allows users to crop or rotate photos as well. Ensure the "Add Photo" button remains accessible to upload

	abandoning the entire listing process.			replacements.
Heuristic 9: Help users recognize, diagnose, and recover from errors	On the Home page, the "Suggested For You" section shows toy recommendations based on browsing history or preferences, but when users encounter items they're not interested in, there's no way to indicate "Not Interested" or "Don't show this type" to improve future suggestions. If the algorithm suggests inappropriate items (wrong age range, categories the user doesn't want), users have no recovery mechanism to correct the recommendations and must repeatedly see irrelevant suggestions.	Suggested For You View all 	Severity: 2 - Minor usability problem Frequency: Less common - affects users who receive poor recommendations, which may be a subset depending on algorithm accuracy. Impact: Moderate impact - causes mild annoyance as users must scroll past irrelevant items, but doesn't prevent them from browsing other sections. Users can still accomplish goals through search and categories.	Add the " Not Interested " option (X button) on each suggestion card that removes it and improves the algorithm. Include feedback prompt: "Why not interested? [Wrong age range] [Don't like this category] [Already have it]" to gather diagnostic data.
Heuristic 5: Error prevention	On the Explore, users can initiate swap requests by simply tapping the "Swap" button without any confirmation dialog or preview of what they're committing to. There's no warning about the exchange requirements. Users could accidentally tap "Swap" while scrolling.		Severity: 2 - Major usability problem Frequency: Common - users browsing quickly might accidentally tap, or new users might not understand swap requirements. Impact: Might frustrate the user every time they miss click the swap button which will lead to another screen Persistence: This issue occurs with every swap request throughout the user's journey.	Add a confirmation dialogue which will pop up when the "swap" button is clicked. If the user agrees to the confirmation, they can be leaded to "swap with" page to proceed further with the swap process

Heuristics Compliances:

Heuristic Rule	Instance of Compliance	Evidence	Justification
Heuristic 1: Visibility of system status	The History page provides clear visual feedback about the status of all user transactions through colour-coded status badges. Users can immediately see which items are "Completed" (green), "Declined" (red), "Pending" (yellow), or "Requested" (yellow), providing continuous awareness of their transaction states.		This design successfully implements visibility of system status by using colour psychology (green=success, red=problem, yellow=in-progress) combined with clear text labels. The semantic colour-coding reduces cognitive load as users can quickly scan the page using colour cues. The date stamps ("15 October 2025," "05 October 2025") provide temporal feedback , helping users track transaction timelines.
Heuristic 2: Match between system and the real world	The Home Page navigation uses universally recognized icons and metaphors that map to real-world concepts: a book for "My List", a location pin for "Near Me", and a heart for "Wishlist". These icons follow established conventions from physical and digital environments.		This exemplifies matching the system to the real world through skeuomorphic design and familiar metaphors . The location pin is universally understood from maps and navigation apps. The heart symbol for favorites/wishlist is a platform-wide standard. Text labels accompany each icon to eliminate ambiguity (redundant encoding). The language is user-centered ("My List," "My Requests") rather than system-centered, making the interface feel personal and natural. This follows natural mapping principles where the relationship between controls and their effects is obvious.
Heuristic 4: Consistency and standards	The bottom navigation bar maintains identical appearance, position, and behaviour across all screens. The five tabs (Home, Explore, Categories, Messages, Donate/Swap) use consistent icons, labels, and the		This demonstrates strong internal consistency (same elements behave identically throughout the app). The tab bar pattern is a standard iOS and Android convention for primary navigation. Icon standardization (home icon, search/explore icon, grid for categories, message bubble, plus sign) follows industry-

	standard highlighted state for the active tab.	 The image shows a navigation bar with a yellow 'Review' button at the top. Below it are five icons: Home, Explore, Categories, Messages, and Donate/Swap.	wide conventions established by major platforms. The active state indication (highlighted/filled icon) provides mode feedback showing the current location.
Heuristic 6: Recognition rather than recall	The List Item Preview page displays all previously entered information in a comprehensive review format, allowing users to recognize and verify their inputs rather than recalling them from memory. All fields (Title, Category, Description, Quantity, Condition, and Safety hazards) are clearly visible with their entered values before final submission.	 The image shows a 'Preview' page for a 'Lego City Set'. It includes fields for Title, Category, Description, Quantity, Condition, and Safety hazards, each with its entered value displayed. There are also photo thumbnails and a 'Add Photo' button.	This design exemplifies recognition over recall by implementing a comprehensive preview pattern before final submission. Users don't need to remember what they entered in previous steps - all information is externalized and visible for review. The visual hierarchy organizes information with clear labels (Title, Category, Description, etc.) that serve as memory cues , reducing cognitive load. The use of pill-shaped tags for Condition and Safety hazards provides visual tokens that are easier to scan and recognize than plain text. The photo thumbnails at the top allow users to recognize their uploaded images without navigating back.
Heuristic 7: Flexibility and efficiency of use	The History page provides efficient filtering through four tab options (All, Donated, Swapped, Requested) that allow experienced users to quickly access specific transaction types, while novice users can view everything in the "All" tab. The "Sort By" feature in the top-right provides additional flexibility for organizing transactions by different criteria (likely date, status, or alphabetical). This dual-layer system accommodates both browsing (novice) and targeted	 The image shows a 'History' page with a back arrow and a 'Sort By' dropdown menu. Below are four tabs: All, Donated, Swapped, and Requested.	This design excellently supports both novice and expert users through progressive filtering. The "All" tab serves novices who want to browse their complete history without making decisions about categorization. Expert users benefit from quick filters - a frequent user who only wants to check swap status can jump directly to the "Swapped" tab, implementing accelerators that bypass unnecessary browsing. The tab pattern requires only one tap to filter, following Fitts's Law with large touch targets that are easy to access repeatedly. The Sort By feature provides additional efficiency for power users who want chronological, alphabetical, or status-based organization, supporting flexible workflows.

	searching (expert) behaviors.		
Heuristic 8: Aesthetic and minimalist design	The Categories page presents information with a clean visual hierarchy, showing only essential elements: category names, representative thumbnail images, and navigation arrows. There's no decorative clutter, redundant text, or unnecessary graphical elements. Each category card communicates its purpose clearly without overwhelming users with options or information.		This exemplifies minimalist design through information hierarchy and reduction. Each category card contains only essential information (name and visual preview), eliminating redundant details like item counts or descriptions that would add noise. The consistent card format creates visual rhythm and scanning efficiency. The thumbnail images provide visual recognition faster than text alone, supporting dual-channel processing. The design follows the principle of proximity by grouping related information (images + category name) within clear boundaries. White space between cards provides visual breathing room, preventing cognitive overload.

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AI Declaration

Use of Generative AI in This Assessment

In accordance with Monash University's policy on the use of Generative Artificial Intelligence (AI) tools, we declare the following use of AI in this submission:

- Generative AI was used only for creating image assets as visual content in our prototype
- All AI-generated content has been properly acknowledged
- No AI tools were used for writing text, or creating any other aspect of this assessment
- All written work, including the report and heuristic evaluations, represents our own original analysis and understanding

UI kits Used:

Kutuku - eCommerce Mobile App UI Kit Figma High Quality Template (Community)

Free UI Icons - Open-Source Vector Icon Set (svg) (Community)

Appendix

Appendix A: Rukma Kuppusamy

2.1 User Persona

MOUNIKA BHARATHI



“Sometimes I look at the mountain of toys my kids have outgrown and I don’t see play – I see waste. I believe in building a community where we share more, but the reality of making it happen feels like another chore.”

AGE 32
OCCUPATION Marketing Coordinator
HOUSEHOLD Married with children
LOCATION Melbourne, VIC
ARCHETYPE The Community Conscious Connector

Mindful

Community-Driven

Sustainability-focused

Clutter-Conscious

Biography

Mounika is the practical yet hopeful heart of her family, juggling a fast-paced marketing career with the joyful chaos of raising two young kids in a compact city apartment. Driven by memories of her own childhood—where sharing leftovers or toys was second nature, she’s determined to pass those values on. Mounika organizes playdates and toy donation drives, quietly inspiring her neighbours to share rather than discard. Still, the reality of limited space and time often clashes with her vision of an eco-friendly, close-knit community. Each week, she feels the tug-of-war between her ideals and the exhaustion of making them real.

Behavioural Trait

SELECTIVE SHARER: Carefully chooses toys to donate or swap, balancing sentiment and utility.

HESITANT REQUESTER: Rarely asks for toys unless she trusts the giver.

COMMUNITY MOTIVATOR: Initiates group swaps, encouraging others to join.

SWAP STRATEGIST: Seeks organised, hassle-free exchanges.

IMPACT-ORIENTED GIVER: Wants to know donations make a real difference.

Motivations & Goals

- Pass on toys thoughtfully, reducing clutter without creating waste or guilt.
- Inspire her children to share, be resourceful, and care for the planet.
- Model eco-friendly, community-minded habits within her family and neighborhood.
- Confidently balance her career, parenting, and sustainability values.
- Make every toy exchange safe, hygienic, and genuinely rewarding for all involved.

Frustrations & Pain Points

- Limited apartment space leading to constant toy clutter.
- Anxiety about hygiene, toy safety, and reliability of strangers.
- The tiring back-and-forth needed to schedule exchanges.
- Constantly juggling work, parenting, and household duties feels overwhelming.
- Her eco-friendly values often clash with practical barriers.

Digital Habits

Mobile-Dependent	
App-Simplicity	
Privacy-Cautious	
Seeks Tech Help	

*AI Generated photo

User Stories

2.1 User Story 1

As Mounika, I want to trust that every toy I borrow or swap is clean and safe, so that I can protect my children’s health and feel secure sharing with others.

MoSCoW Priority: Must Have

2.2 User Story 2

As Mounika, I want to easily keep track of all the toys I’ve shared, swapped, or received, so that I can manage my clutter and avoid misunderstandings with others.

MoSCoW Priority: Should Have

Functional User Requirements

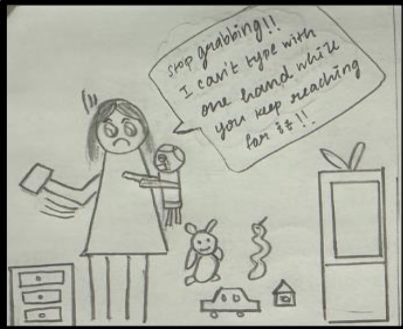
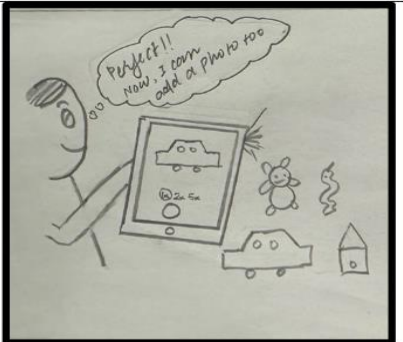
2.3.1 User requirement 1

Users should be able to view verified cleanliness and safety information for every toy before borrowing or swapping.

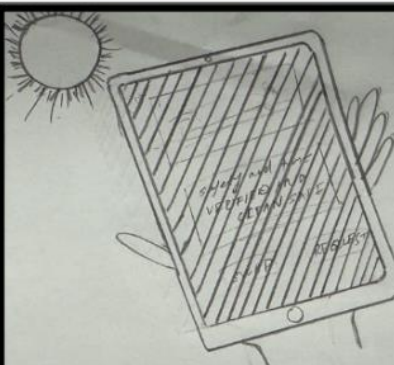
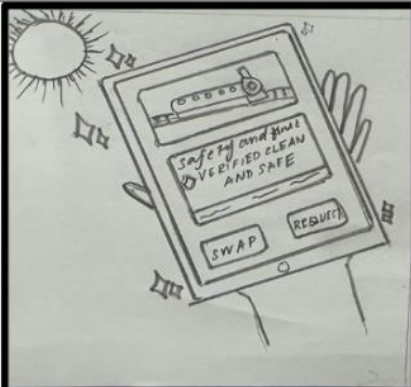
2.3.2 User requirement 2

Users should be able to access and manage a personal dashboard that tracks all toys they have shared, swapped, or received.

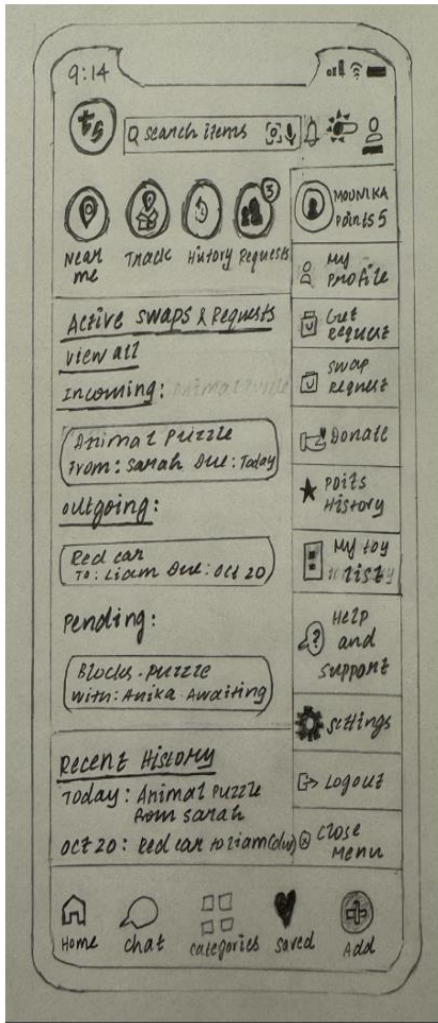
Storyboard -1

Persona : Mounika Bharathi	Scenario: Mounika needs to log toys in ToySwap but has only one hand free while holding her baby on her hip.	
User Story: <i>As Mounika, I want to easily keep track of all the toys I've shared, swapped, or received, so that I can manage my clutter and avoid misunderstandings with others.</i>		
Accessibility & Inclusivity Focus: One-handed use while holding a baby on her hip (situational motor impairment).		
		
Mounika stands in her living room holding her baby on her hip, looking at the pile of toys scattered on the floor. She needs to log which toys are for sharing but worries she'll forget if she doesn't do it now.	Mounika tries to use her phone to log the toys, but her baby keeps grabbing at it, making it impossible to type with one hand. Frustrated, she realizes this traditional method won't work while juggling parenting duties.	Mounika suddenly remembers ToySwap has voice control. She places her phone on the table and speaks: "Hey ToySwap, add the red toy car to my sharing list."
		
Using ToySwap's camera function, Mounika takes a photo of the red toy car to add to her listing. With the baby calm and the hands-free voice entry complete, she can now document her toys easily.	While holding her baby with both hands, Mounika hears an audio notification from her phone on the table: "Confirmed: Red Toy Car added to sharing list." She stays informed without needing to look at or touch her phone.	Later, Mounika's living room is tidy with all toys organized. She plays happily with her baby, ToySwap's voice control and audio notifications helped her manage her toy sharing commitments effortlessly, even with her hands full.

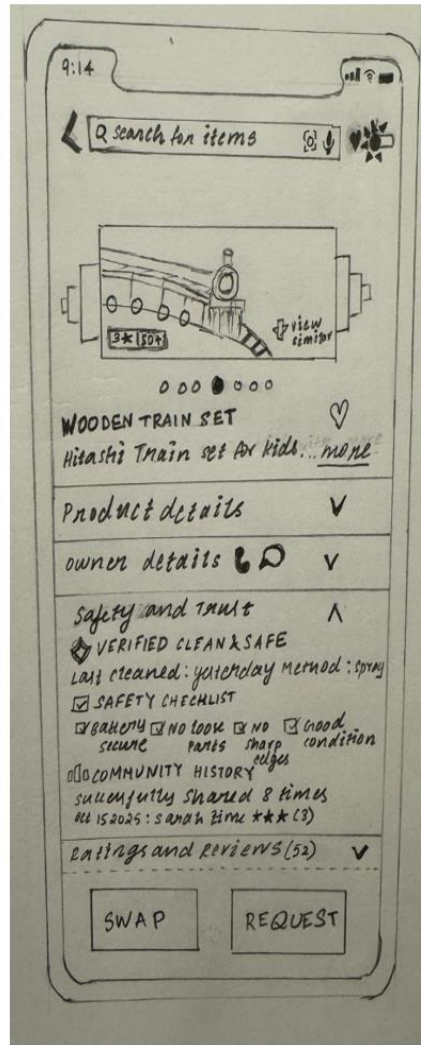
Storyboard -2

Persona : Mounika Bharathi	Scenario: Mounika has diabetic retinopathy and is trying to use ToySwap in bright sunlight to verify if her kids' toys are safe and verified.	
User Story: <i>As Mounika, I want to trust that every toy I borrow or swap is clean and safe, so that I can protect my children's health and feel secure sharing with others.</i>		
Accessibility & Inclusivity Focus: Permanent visual disability (diabetic retinopathy) + Situational visual impairment (bright sunlight)		
		
Mounika sits on a park bench in bright sunlight, trying to check toy safety information on ToySwap while watching her children play. Despite her glasses, the screen is washed out and she struggles to read it.	The low-contrast interface combined with screen glare makes text and safety badges nearly impossible to read. Mounika squints at the indistinct information, unable to verify if the toy is safe for her children.	Mounika notices a "High Visibility Mode" button with a sun icon in the corner of the app. Hopeful for a solution, she taps it to see if it will help her see the screen better.
		
The screen instantly transforms to a dark background and extra-large "VERIFIED CLEAN & SAFE" badges. Mounika can finally read all the information clearly, even in the bright sunlight.	With the high visibility interface, Mounika easily taps the large "Request" button with one hand while keeping an eye on her children playing. She feels reassured seeing the clear safety verification.	Mounika puts her phone away confidently. The High Visibility Mode remains active and works with her other assistive technologies, giving her independence and the ability to make informed decisions about her children's toys.

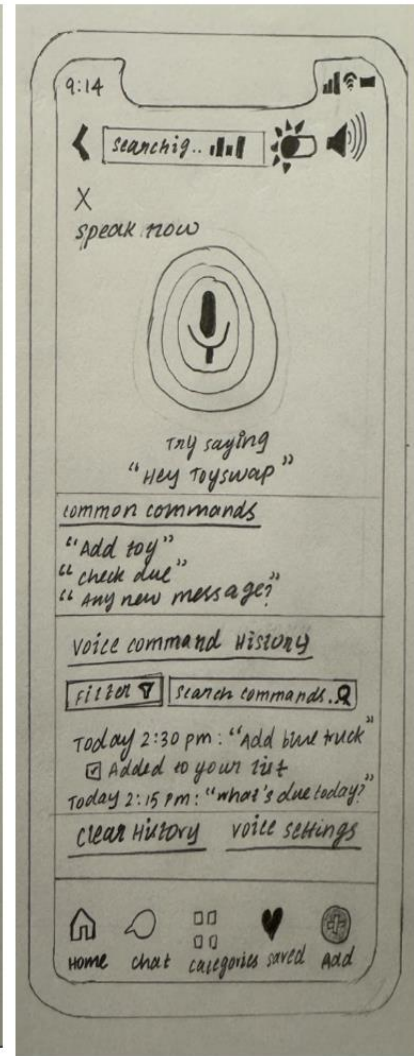
Low-Fidelity Prototype



Screen 1: Dashboard



Screen 2: Toy Detail Screen



3: Voice Control Interface

Appendix B: Rachit Rajiv

2.1 User Persona

Alex Warner



Biography

Alex loves hosting game nights in his Clayton sharehouse. He loves the social connection of board games but feels the pressure to always provide something new and exciting for his friends. His part-time job means every purchase is a careful calculation. He's tech-savvy but has been burned by no-shows on marketplace apps before. He wants to be sustainable but feels guilty that his 'to-play' pile of games is gathering dust.

Motivations & Goals

- Access a rotating library of games without the financial commitment.
- Be the host with the most (recent games) without going broke.
- Declutter his space and find a new home for his old games easily.
- Feel confident that any game he gets is complete and in good condition.
- Discover new games that match his and his friends' interests reliably.

Frustrations & Pain Points

- High-quality board games are a significant investment on a student budget.
- His sharehouse is small. His growing collection is starting to annoy his roommates.
- Doesn't always know what to do with old or unused games.
- Listing items for sale online is tedious.
- A lack of trusted and efficient platforms for borrowing and swapping games.

Digital Habits

Smartphone-Dependent

App customisation

Privacy-Cautious

Accessible Processes

Behavioural Trait

SOCIAL CONNECTOR: Naturally, it takes on the role of community builder and host.

BUDEGT CONSCIOUS: Sensible with money and value conscious.

TECH SAVVY: Digitally native and comfortable with technology.

SPACE AWARE: Highly conscious of his physical environment's limitations.

Quote:

"I'd love to try the new trends, but I am still broke from the last one. A try-before-you-buy swap would be perfect."

AGE 21

OCCUPATION Full time student

HOUSEHOLD Shared accomodation

LOCATION Clayton ,VIC

ARCHETYPE The Community Conscious Connector

Resourceful **Social**

Pragmatic **Tech savvy**

**AI Generated photo*

User Stories

2.1 User Story 1 (Must Have)

As Alex, a budget-conscious student, I want to easily browse games with clear condition details, so that I can swap confidently without worrying about hidden issues or wasting money.

2.2 User Story 2 (Should Have)

As Alex, who values safe and reliable interactions, I want a secure way to arrange swaps with other users, so that I can reduce no-shows and feel more in control of the process.

Functional User Requirements

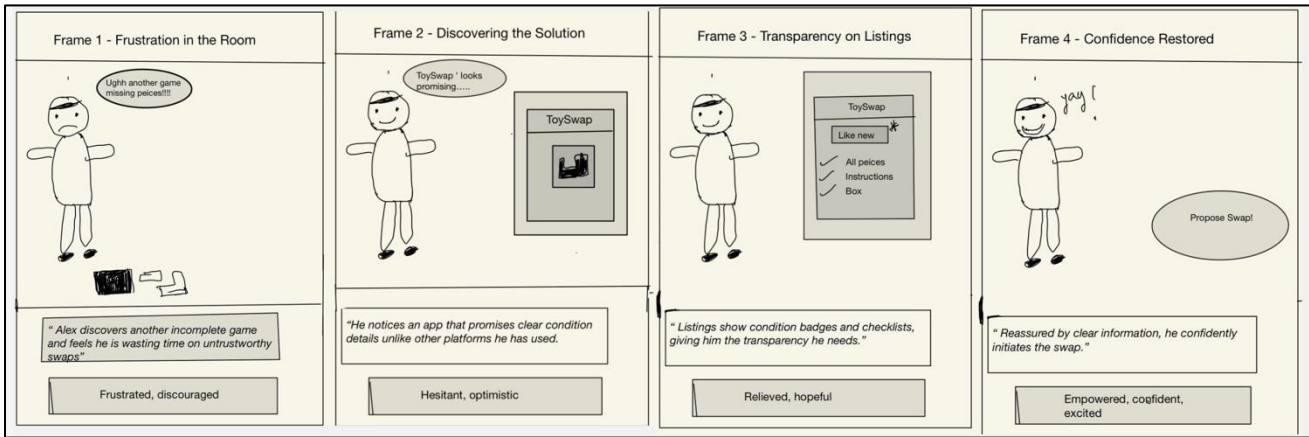
2.3.1 User requirement 1

Users should be able to browse and filter available items using detailed condition tags selected from a standardized checklist

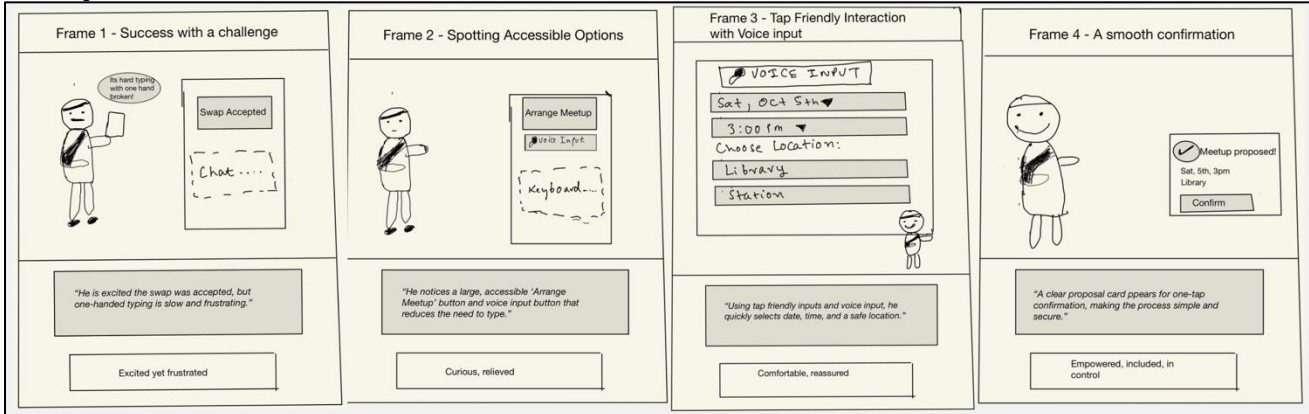
2.3.2 User requirement 2

Users should be able to communicate with others through secure, in-app messaging that doesn't expose personal contact details

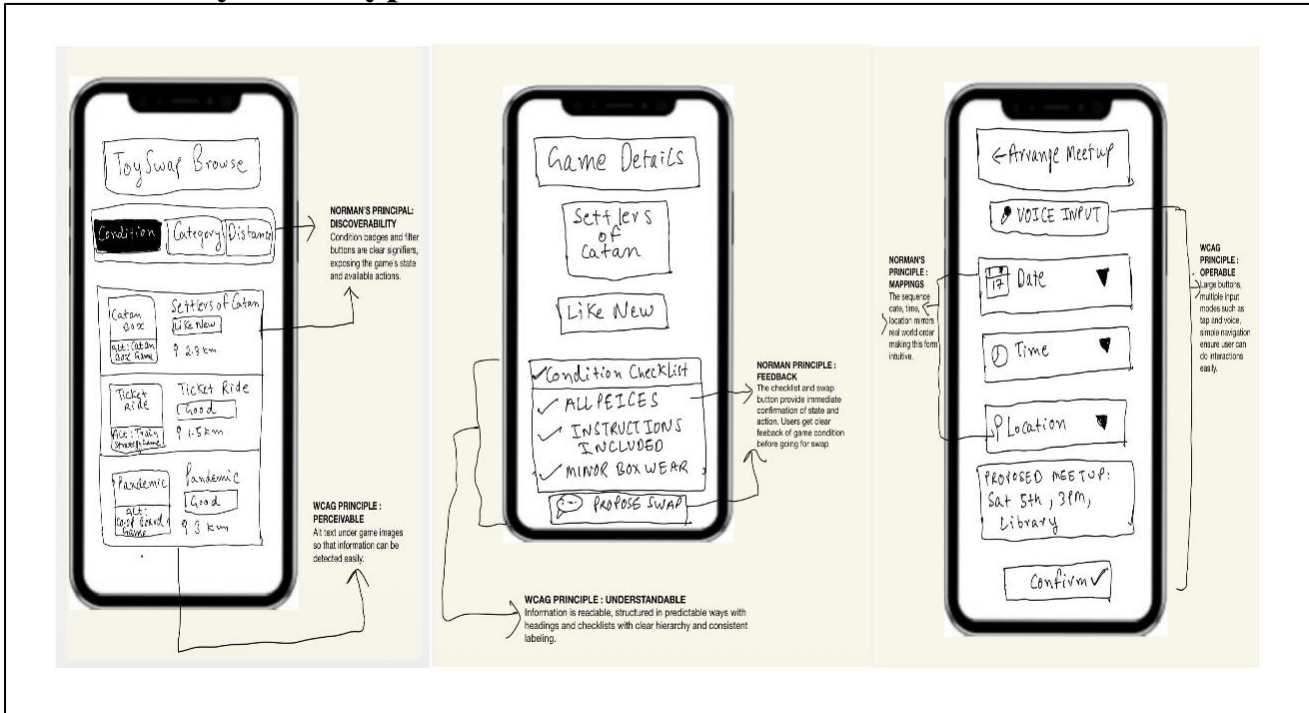
Storyboard -1



Storyboard -2



Low-Fidelity Prototype



Appendix C: Rishabh Ray

2.1 User Persona

ARJUN MEHTA



“Every child deserves safe, joyful play — no matter where the toy comes from.”

AGE 32
OCCUPATION Childcare Centre Founder
HOUSEHOLD Married with children
LOCATION Melbourne, VIC
ARCHETYPE Community Builder

Practical Sustainability-Focused
Child-Safety Advocate Community-Driven

Biography

Arjun is the founder of a small childcare initiative in Melbourne. As a father himself, he knows the importance of safe, engaging toys for children's growth. Rising childcare costs and limited resources pushed him to find creative solutions — leading him to bulk toy donations.

For Arjun, pre-loved toys aren't second-hand, they're opportunities: affordable, sustainable, and meaningful when given new life in a child's hands. He actively connects with local parents, schools, and online donation groups, often juggling the dual roles of organiser and caregiver. Trust is central to his work — every toy must be safe, clean, and worthy of a child's playtime.

Behavioural Trait

SAFETY-FIRST: Checks every toy before use, prioritising hygiene and safety.

RESOURCEFUL COLLECTOR: Proactively seeks bulk donations to stretch limited funding.

CHILD-CENTRIC THINKER: Selects toys based on educational and developmental value, not just availability.

VARIETY SEEKER: Values having a wide range of toys to match children's learning needs.

Motivations & Goals

- Secure bulk toy donations to reduce costs and ensure continuous supply.
- Ensure all donated toys meet hygiene and safety standards.
- Build trust with parents and donors.
- Provide a diverse range of play and learning opportunities.
- Promote sustainability by giving pre-loved toys a second life.

Digital Habits

Community-Platform User:

Visual Communicator:

Tech Explorer:

Privacy Awareness:

Frustrations & Pain Points

- Difficulty verifying cleanliness and safety of donated toys.
- Limited funding to purchase toys new.
- Inconsistent supply of donations in bulk.
- Time constraints from balancing childcare management and donation sourcing.
- Rising demand from families can sometimes exceed available resources.

**AI Generated Image*

User Stories

Story 1 – Must Have

As Arjun, I want to collect safe and hygienic toys in bulk so that I can provide enough play and learning materials for all children in my childcare centre.

Story 2 – Should Have

As Arjun, I want to build trust with parents and donors so that I can consistently receive reliable toy contributions for my childcare centre.

Functional User Requirements

2.3.1 User requirement 1

Users should be able to contribute toys in bulk with clear details about quantity, condition, and age suitability.

2.3.2 User requirement 2

Users should be able to view and verify donation history to build trust and transparency between donors and recipients.

Storyboards

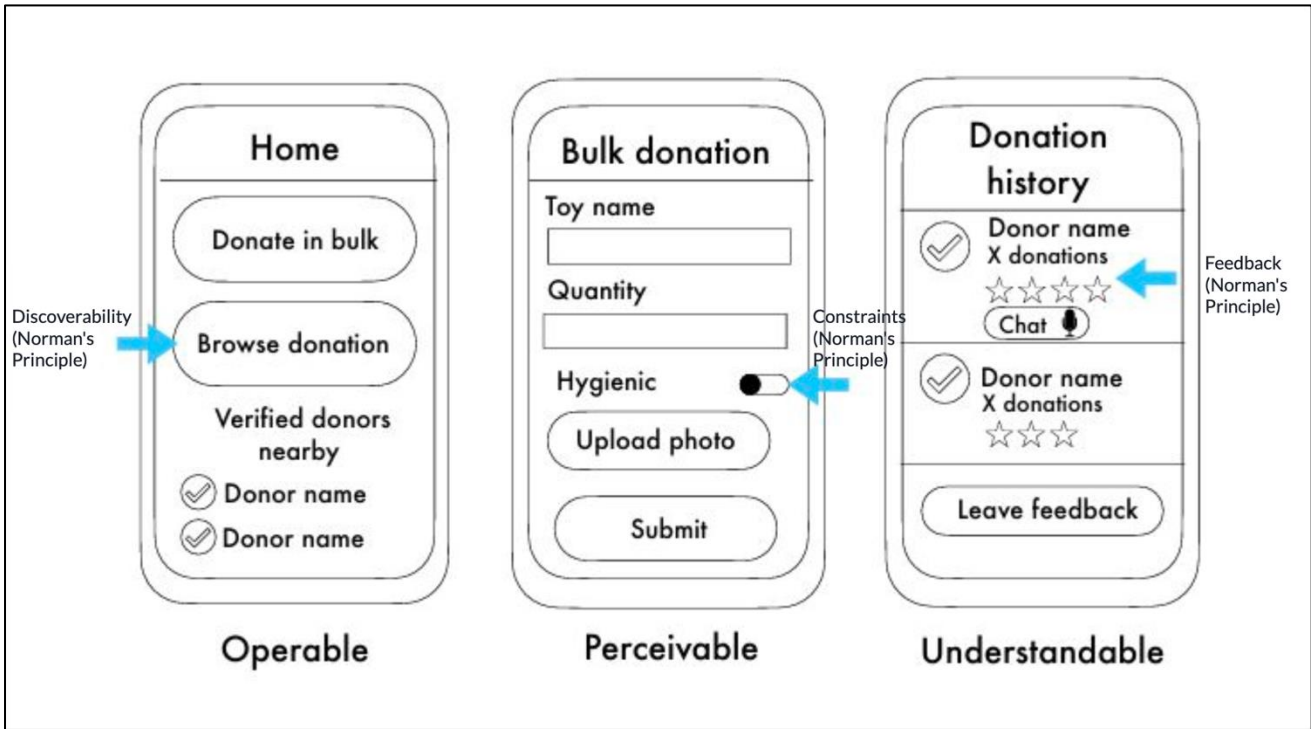
Storyboard A – Bulk Safe Donations (Story 1)



Storyboard B – Trust with Donors (Story 2)



Low-Fidelity Prototype



Appendix D: Anish Srinath

2.1 User Persona

Aarav Sharma



"I want to declutter before moving cities and donate my toys so children who can't afford them can enjoy and learn."

AGE 23
OCCUPATION Software Developer
HOUSEHOLD Shared Household
LOCATION Melbourne
ARCHETYPE Time-Pressed Trust-Seeking Donor

Community Driven Clutter-Conscious
Impact Oriented Trust Seeker

Biography

Aarav has just completed his Bachelor's in Computer Science from Monash University, Melbourne. He is now moving to Sydney for his first full time job as a Software Developer. As part of the move, he wants to clear the clutter of childhood belongings, especially toys and board games. He strongly believes these items should benefit children who cannot afford them, rather than sitting unused. Aarav values community giving and meaningful contribution as he starts his independent life

Behavioural Trait

- IMPACT-VERIFIED GIVER:** Donates only when he can see proof the items reach real beneficiaries.
- TIME-PRESSED DECLUTTERER:** Acts in short bursts before moving; prefers bulk upload and quick drop-offs.
- LOW-FRICTION NAVIGATOR:** Chooses apps with simple flows, few settings, and clear CTAs.
- TRUST-BADGE SEEKER:** Looks for verified profiles, ratings, and transparent handover details.

Motivations & Goals

- Donate unused toys to children who need them
- Make moving easier by reducing clutter
- Support sustainability through reuse
- Find simple, trustworthy donation channels
- Help Others
- Reduce Waste and promote reuse
- Believes in contributing and giving it back to the community
- Start a "fresh chapter" without baggage

Frustrations

- Unsure where to donate toys effectively
- Concern that donations may not reach real beneficiaries
- Limited time due to relocation and job start
- Finds most apps/platforms too complicated or not transparent

Digital Habits

Phone-first usage	100%
Simple-UI Preference	100%
Permission Cautious	70%
Self-Help with Tech	50%

*AI Generated Image

User Stories

User Story 1

As Aarav, I want to quickly understand how to donate items without getting lost or frustrated, so that I can successfully contribute to my community without abandoning the process.

User Story 2

As Aarav, I want to understand what information the app needs from me and why, so that I can feel confident about my privacy while still accessing location-based donation opportunities.

Functional User Requirements

2.3 New Functional User Requirements

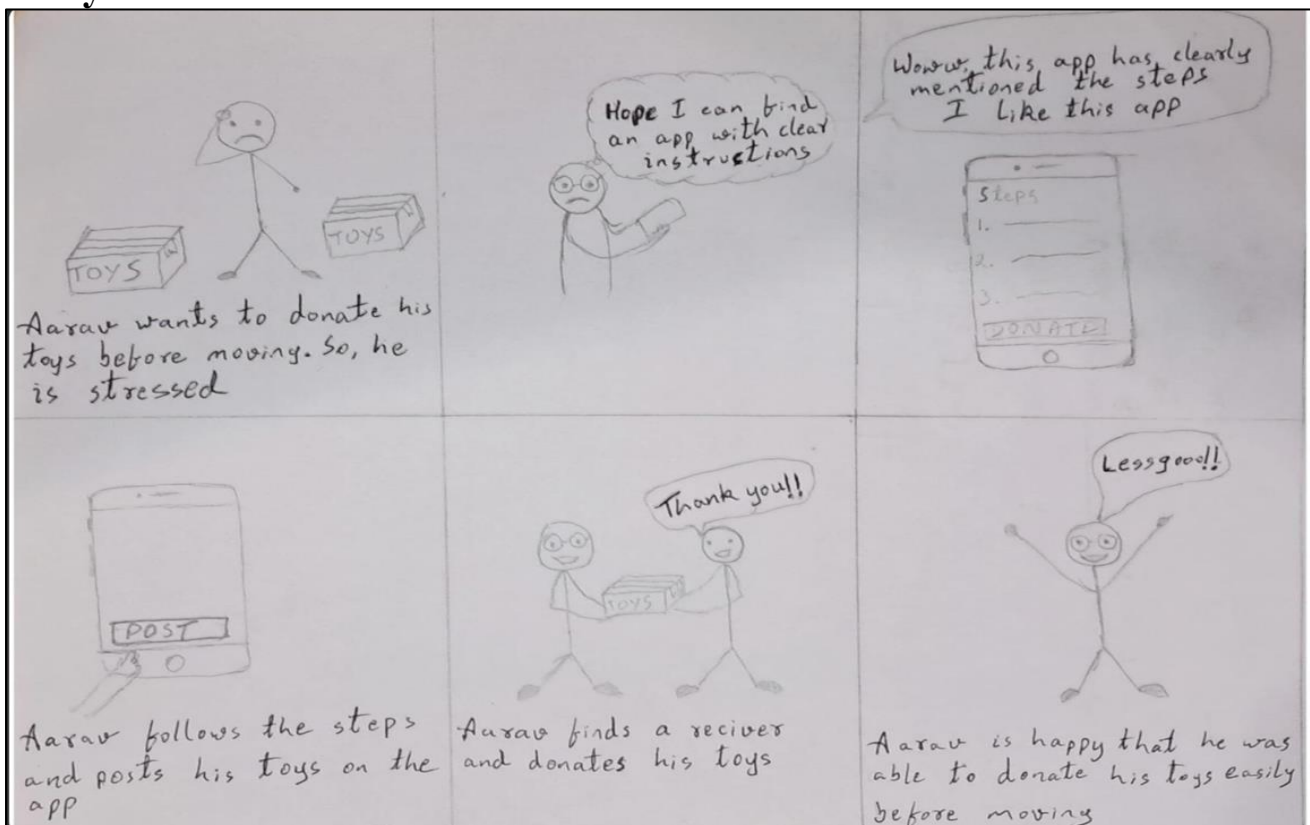
2.3.1 User requirement 1

Users should be able to access key actions (such as finding, donating, and swapping items) through clearly labelled navigation options.

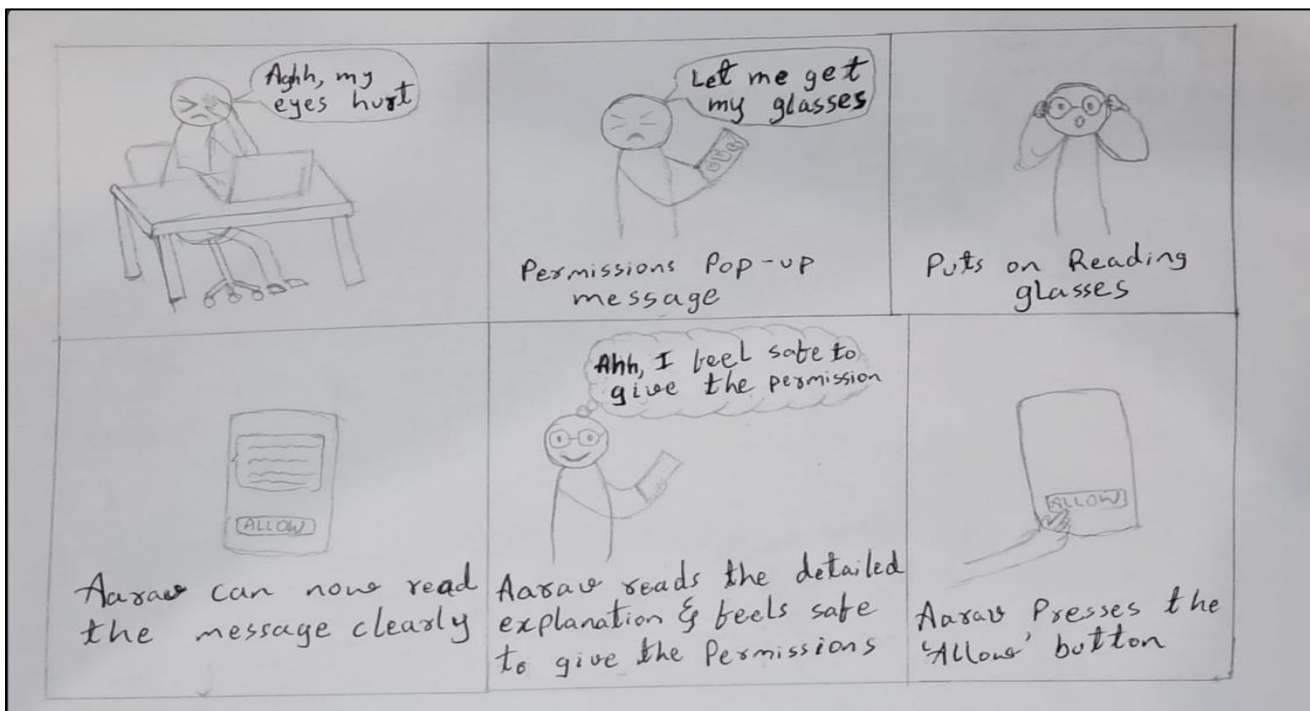
2.3.2 User requirement 2

Users should be able to view clear explanations when the app requests permissions (e.g., location, camera).

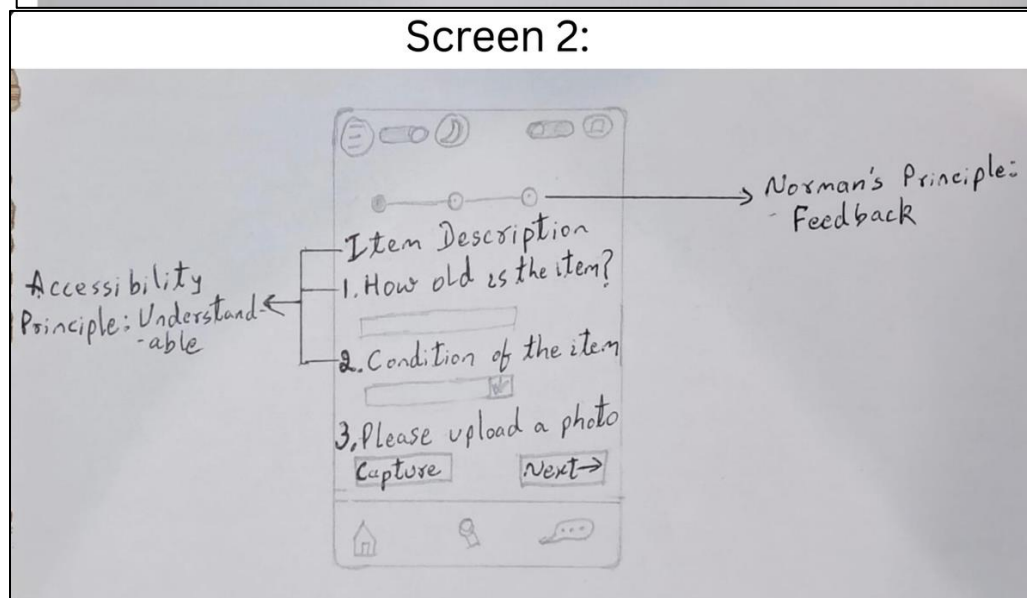
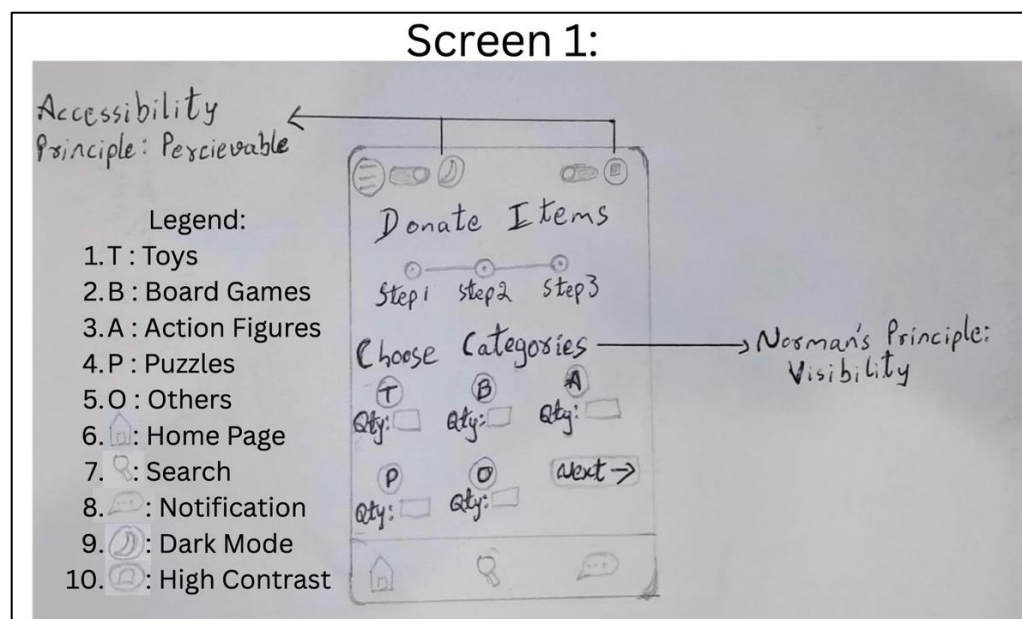
Storyboard-1



Storyboard-2



Low-Fidelity Prototype



Screen 3:

